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Supplementary information

Inverted perovskite solar cells using dimethylacridine-based dopants

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Supplementary Materials for

Inverted perovskite solar cells using dimethylacridine-based dopants

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Materials and Methods

Materials**:** Anhydrous solvents including *N*, *N*-dimethylformamide (DMF), dimethyl sulfoxide (DMSO), chlorobenzene (CB), isopropanol (IPA), tetrahydrofuran (THF) and Cesium iodide (CsI, 99.999%) were purchased from Sigma-Aldrich. Lead iodide (PbI_2, 99.99%) and lead bromide ($PbBr_2$, 99.99%) were purchased from TCI. Formamidinium iodide (FAI, 99.99%) and methylammonium bromide (MABr, 99.99%) were purchased from Great Cell Solar. Methylammonium chloride (MACl, 99.9%), bathocuproine (BCP, >99%), phenyl-C_{61}-butyric acid methyl ester ($PC_{61}BM$, >99%) and benzeneethanamine, hydrobromide (PEABr, >99.5%) were purchased from Lumtec. 9,9-Dimethyl-9,10-dihydroacridine (98%), tetrabutylammonium bromide (98%), N-bromosuccinimide (98%), dibromobutane (98%), triethyl phosphite (97%) were purchased from the Energy Chemical Co. Ltd. Chloroform-d and methyl sulfoxide-d6 were purchased from the J&K Scientific. ITO glass were purchased from Suzhou ShangYang Solar Technology Co. Ltd. All materials above were used as received. (4-(2,7-dibromo-9,9-dimethylacridin-10(9H)-yl)butyl)phosphonic acid (DMAcPA) was synthesized in our lab, the detailed synthetic procedure was shown in the part of synthesis.

***Preparation of pristine and molecule-doped perovskite precursor solution**:* 6.7mg MABr, 8.8 mg MACl, 18.2 mg CsI, 196.1 mg FAI, 22 mg $PbBr_2$ and 572 mg PbI_2 were dissolved in 865 μl DMF and DMSO solution (volume ratio 4:1) and stirred at 60 ^{o}C for 1 h to prepare undoped precursor solution. For doped solution, another 1~7 mg DMAcPA was added in the preceding solution.

Device fabrication: The inverted (p-i-n) perovskite solar cell studied in this paper with an architecture of ITO/Perovskite/$PC_{61}BM$/BCP/Ag. Depending on the ITO/Perovskite(DMAcPA) Schottky contact, we built the inverted device with its device structure shown as the inset in Fig. 5A. The ITO/DMAcPA/Perovskite is taken as the control to evaluate the routine inverted device with DMAcPA as the hole selection layer[16-18]. The ITO glass substrate was cleaned successively with detergent (2% in deionized water), deionized water, acetone and isopropanol, each for 15 minutes in an ultrasonic bath. After that, the substrate was dried with N_2 and treated with UV-ozone (15 min). For control devices, DMAcPA with a concentration of 0.5~3 mg/mL in THF were spin coated on the cleaned ITO substrates and annealed at 100 ℃ for 10 min. The as-prepared perovskite precursor solution was spin-coated onto the modified ITO substrate with speed of 1000 rpm for 5 s and 4000 rpm for 30 s. During the last 15 s of the spinning process, the liquid film was treated by drop-casting chlorobenzene solvent (150 μl). For target device, the doped solution was spin-coated on cleaned ITO substrates directly. The substrates were annealed on a hot plate at 100 ℃ for 50 min, and PEABr with concentration of 1 mg/ml in IPA and DMSO solution (volume ratio 200:1) was dynamically spin-coated on top (4000 rpm, 30 s) and annealed at 100 ℃ for 5 min. Then a layer of $PC_{61}BM$ (20 mg/ml in chlorobenzene, 1500 rpm, 35 s) was coated and annealed at 100 ℃ for 10 min, followed by spin-coating BCP with concentration of 0.5 mg/ml in IPA on top (4000 rpm, 30 s). Finally, 100 nm thick of Ag was deposited on top by thermal evaporation.

***Materials characterizations*:** NMR spectra were recorded on Bruker Avance 500 and Bruker Avance 400 spectrometer in $CDCl_3$,DMSO-d_6, CD_2Cl_2 and DMF-d_7. High resolution mass spectra (HRMS) were recorded on Thermo Scientific Q Exactive mass spectrometer, using electro spray ionization (ESI) as ionization mode.Thermal gravimetric analysis (TGA) was recorded on Mettler TGA1 with a heating rate of 10 ℃/min under nitrogen flow. UV-vis spectra were recorded on a PerkinElmer Lambda 950 spectrometer. The steady state and time resolved PL spectra were measured using a Spectrofluorometer (FS5, Edinburgh instruments) and 405 nm pulsed laser was used as excitation source for the measurement.

Films characterization: X-ray diffraction (XRD) spectra were measured by Bruker D8 ADVANCE. X-ray photoelectron spectroscopy (XPS) measurements were carried out on an Omicron ESCA Probe XPS spectrometer (Thermo Scientific ESCALAB 250Xi) using 150 eV pass energy and 1 eV step size for survey scan and 20 eV pass energy and 0.01 eV step size for the fine scan. Contact angle measurements were carried out with a KLA-Tencore D120 surface profiler. SEM was characterized by HITACHI SU8230. FTIR was characterized by Bruker Vertex 70v.

Atomic force microscopy-based infrared spectroscopy (AFM-IR): AFM-IR was characterized by nanoIR2 system (Anasys instruments) and the stretching vibration of P-C at 1410 cm^{-1} was used to define DMAcPA.
In addition to the main text discussion, Fig. 2A-C show the AFM surface morphologies of the top, bottom and detached ITO of the DMAcPA/Perovskite film. Fig. 2D-F are the molecule plenty distribution images corresponding to the regions of Fig. 2A-C, respectively, which are in the same plenty bar and herein show the clear contrast. Brighter yellow means higher plenty of DMAcPA while the red denotes little.The latter can be due to the strong coordination of Pb--O. For the Perovskite(DMAcPA) film, the molecule plenty distribution images of Fig 2J-L are corresponding to the morphologies of Fig. 2G-I. Its detached ITO surface is covered with more DMAcPA molecules than that of the DMAcPA/Perovskite. Regarding the Perovskite(DMAcPA), that means DMAcPA molecules always stick firmly to the PbI_2 units in the perovskite crystallization process from top to bottom.Another interesting point is that the top surface of the Perovskite(DMAcPA) film also contains a certain amount of DMAcPA at the grain boundaries, in comparison of the blank background of the DMAcPA/Perovskite top surface.

Time of Flight Secondary Ion Mass Spectrometry (ToF-SIMS):Depth profiling data were obtained with ToF-SIMS 5 system from ION-TOF. Bi^{3+} ions were used as primary ions and negative ions were detected. Sputtering was performed using Cs^+ sputtering ions with 500 eV ion energy, 50 nA ion current and a 200×200 μm^2 raster size. An area of 50×50 μm^2 was analyzed using Bi^{3+} ions with 30keV ion energy.

To verify the above point, time-of-flight secondary ions mass spectroscopy (ToF-SIMS) was employed to characterize the distribution of DMAcPA throughout the perovskite film. Fig. 2M shows the in-depth PO_2^- distribution of the ITO/DMAcPA/Perovskite film along with PbI^- and In^- as references. For the undoped PVK, no PO_2^- signal of DMAcPA turns up, which surges up sharply when encountering the pre-coated DMAcPA layer on ITO. In a clear contrast, for the ITO/Perovskite(DMAcPA) film (Fig. 2N), the PO_2^- signal of DMAcPA exists throughout the doped PVK layer and rises sharply too at the bottom. Moreover, as the onset of the blue line shows, that sharp rise of PO_2^- signal occurred after a delay compared with the ITO/DMAcPA/Perovskite, which means the layer of the automatically extruded DMAcPA-PbI_x agglomerates is thinner than the pre-coated DMAcPA layer on the bare ITO. The ToF-SIMS data unambiguously examine again the conclusion drawn from both SEM and AFM-IR images that DMAcPA exist at both the grain boundaries throughout the doped perovskite film and the bottom interface with ITO.

Device performance characterization: J-V measurements were carried out using a Keithley 2400 source meter in ambient environment (~20 ℃ and ~60% RH). The unencapsulated devices were measured both in reverse scan (1.2 V→-0.2 V, step 0.01 V) and forward scan (-0.2 V→1.2 V, step 0.01 V) with 200 ms delay time. Illumination was provided by an Oriel Sol3A solar simulator with AM 1.5G spectrum and the light intensity of 100 mW/cm^2 was calibrated by a standard KG-5 Si diode. The active area (0.08 cm^2) of our device was calibrated with a shadow mask during the measurements. EIS and Mott–Schottky plots were conducted on the Zahner Zennium electrochemical workstation. EQE measurements for devices were conducted with an Enli-Tech (Taiwan) EQE measurement system, and the light intensity at each wavelength was calibrated with a standard single-crystal Si photovoltaic cell. For the stabilized power output (SPO) tests under illumination, bias voltage was fixed at the maximum power point voltage (V_{max}), and the current density variation in ambient environment was recorded without controlling the device temperature. The long-term operational stability was conducted by applying the encapsulated devices under 1 sun equivalent LED lamp in ambient environment (~25 ℃ and ~50% RH). The maximum power point was recorded every minute.

Ultraviolet photo-electron spectroscopy (UPS): To measure work function, ultraviolet photoelectron spectroscopy (UPS) spectra were recorded with an Imaging Photoelectron Spectrometer (Axis Ultra, Kratos Analytical Ltd), with a non-monochromated He Iα photon source (h_ν = 21.22 eV). Au is taken as a reference.

Deep-level transient spectroscopy (DLTS): The DLTS measurement was performed via a Phystech FT-1230 HERA-DLTS system. The modified Boonton 7200 capacitance meter (1–75 kHz) was used to examine dynamic capacitance. The effective test area is about 0.14 cm^{-2}. Then the samples were placed in a liquid-helium cryostat (Lakeshore 335,336). The DLTS temperature scan range was from 370 to 205 K at 2 K intervals. The Arrhenius equation for the hole traps is,

$$\ln\left(\tau_e v_{th,\ p} N_V\right) = \frac{E_T - E_V}{k}\frac{1}{T} - \ln(X_h \sigma_h)$$

Here, τ_e is the emission time constant; $v_{th,\ p}$ is the thermal velocity; N_V is the effective density of states of holes; E_T is the energy level of the trap center; E_V is the energy level of VBM; X_h is the entropy factor for holes; σ_h is the capture cross-section for holes.

Transient absorption (TA) spectroscopy: The femtosecond TA spectra were characterized by a setup based on a regenerative amplified Ti:sapphire laser system from Coherent (800 nm, 35 fs, 6 mJ pulse-1, and 1 kHz repetition rate), nonlinear frequency mixing techniques and the Femto-TA100 spectrometer (Time-Tech Spectra).

pH measurements

The pH of precursor solution was collected using Rex Electric Chemical E-301-QC pH meter. The pH meter was calibrated with three standard solutions before each test. After calibration, the pH count was washed several times with the pure solvents. Then, each solution was tested.

Hall measurements

Hall measurements were performed on HL5500 Hall system from Nanometrics Co. Ltd. at room temperature and dark conditions. Samples were prepared in Van der Pauw geometry. Connections of silver wires for the electrical measurements were made by conductive silver paste. After drying process of the paste, the ohmic behavior of the metal contacts was S4 checked by the linear variation of the current-voltage (I–V) characteristics which was observed to be independent of the polarity of the applied current and the contact combinations for each sample. Measurements were done under 0.5T magnetic field. The sample size was 15×15 mm^2. The DMAcPA doped perovskite is tested to be of positive Hall coefficient and herein p-type, which is consistent to the UPS test results discussed in our work.

Density functional theory (DFT) calculation: All the first-principles calculations were carried out based on the density functional theory (DFT), as implemented in the Vienna Ab-initio Simulation Package *(1)*. The Perdew-Burke-Ernzerhof form of exchange-correlation functional (*2*) were utilized. The plane-wave energy cutoff was 290 eV, together with the projector augmented wave pseudopotentials. A single gamma point was used for the *k*-point sampling for molecular systems and a $1 \times 2 \times 1$ *k*-point mesh for the slabs. The structures were optimized with an energy and force tolerance of 10^{-6} eV and 0.02 eV/Å, respectively. The surface adsorption was carried out on a PbI_2-terminated $FAPbI_3$ slab with a size of $19.2 \times 12.8 \times 11.3$ Å^3, together with a vacuum layer of 21 Å in the out-of-plane direction. The binding energy E_b of is defined by $E_b = E(\text{complex}) - \sum E(\text{component})$. The Van der Waals interactions in the dimers are simulated using the DFT-D3 approach (*3*). All the DFT calculations were carried out on the Taiyi cluster supported by the Center for Computational Science and Engineering of Southern University of Science and Technology.

Chemical structure and synthesis of DMAcPA

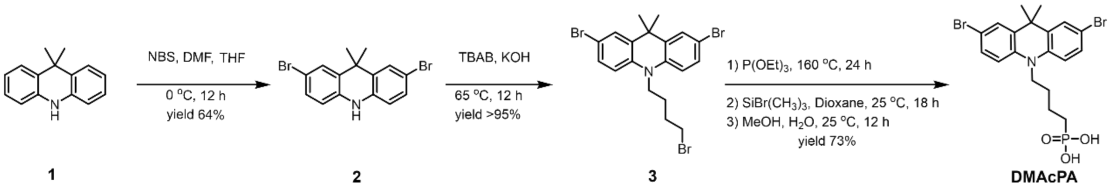


Figure S1. The chemical structure and synthesis procedure of DMAcPA.

Synthesis of 2,7-dibromo-9,9-dimethyl-9,10-dihydroacridine (**2**)

A 100 mL round bottom flask was charged with 9,9-dimethyl-9,10-dihydroacridine (**1**) (0.89 g, 4.24 mmol) and dichloromethane (10 mL) in ice bath, then N-bromosuccinimide (1.51 g, 8.48 mmol) was added in portion. The mixture was warmed up to room temperature and stirred overnight. The reaction was quenched with water and extracted with dichloromethane, the organic layers were combined and dried with anhydrous magnesium sulfate, then removed the organic solvent with rotary evaporator to give the crude product. A flash column chromatography with eluent as petroleum ether/dichloromethane of 10/1 was used to further purification. The title compound **2**was white solid (1.40 g, yield 90%).

^{1}H NMR (400 MHz, DMSO-d_6) δ 9.17 (s, 1H), 7.45 (d, J = 2.3 Hz, 2H), 7.21 (dd, J = 8.5, 2.2 Hz, 2H), 6.73 (d, J = 8.5 Hz, 2H), 1.46 (s, 6H) ppm.

^{13}C NMR (101 MHz, DMSO-d_6) δ 137.99, 130.54, 130.02, 128.64, 115.99, 111.54, 36.59, 31.42.

Synthesis of 2,7-dibromo-10-(4-bromobutyl)-9,9-dimethyl-9,10-dihydroacridine (**3**)

A 100 mL two neck flask was charged with intermediate**2** (1.00 g, 2.72 mmol) and tetrabutylammonium bromide (0.32 g, 0.27 mmol)were dissolved in dibromobutane (15 mL), then 50% KOH aqueous solution (5 mL) was added dropwise. The mixture was heated to 65 ℃ and stirred overnight. The reaction was quenched with water and extracted with dichloromethane, the organic layers ware combined and dried with anhydrous magnesium sulfate, then removed the organic solvent with rotary evaporator to get the crude product. Silica gel column chromatography was used for further purification with eluant as petroleum ether/dichloromethane = 10/1 to give the title compound (**3**) as colorless oil (1.20 g, yield 89%).

^{1}H NMR (400 MHz, $CDCl_3$) δ 7.48 (d, J = 2.3 Hz, 1H), 7.32 (dd, J = 8.7, 2.3 Hz, 1H), 6.82 (d, J = 8.8 Hz, 1H), 3.97 – 3.83 (m, 1H), 3.48 (t, J = 6.1 Hz, 1H), 2.14 – 1.90 (m, 1H), 1.50 (s, 3H) ppm.

^{13}C NMR (101 MHz, $CDCl_3$) δ 139.29, 134.15, 129.57, 127.55, 114.21, 113.42, 45.05, 36.57, 32.84, 30.12, 28.57, 24.44 ppm.

Synthesis of (4-(2,7-dibromo-9,9-dimethylacridin-10(9H)-yl)butyl)phosphonic acid (**DMAcPA**)

A 100 mL Schlenk tube was charged with intermediate **3** (1.20 g, 2.39 mmol) and triethyl phosphite (10 mL), then the mixture was heated to 160℃ and stirred overnight under nitrogen atmosphere. The extra triethyl phosphite was removed by reduced pressure distillation to give a light-yellow oil without further purification. The light-yellow oil was dissolved in anhydrous 1,4-dioxane (10 mL) at room temperature and trimethylbromosilane (3.66 g, 23.90 mmol) was added dropwise, then the mixture was stirred overnight. The 1,4-dioxane was removed with rotary evaporator to give a white solid. The white solid was dissolved in methanol (10 mL) at room temperature, then deionized water was added dropwise until the mixture became opaque and stirred for another 12 h. The crude product was collected by filtration and washed with deionized water. The crude product was dissolved in THF (5 mL) andreprecipitate in acetone (20 mL) and filtrated to give the final product as white solid (0.94 g, yield 78%).

^{1}H NMR (400 MHz, DMSO-d_6) δ 7.48 (d, J = 2.4 Hz, 2H), 7.33 (dd, J = 8.8, 2.3 Hz, 2H), 7.03 (d, J = 8.9 Hz, 2H), 3.89 (t, J = 7.6 Hz, 2H), 1.74 (td, J = 9.7, 7.2, 3.1 Hz, 2H), 1.66 – 1.54 (m, 4H), 1.42 (s, 6H) ppm.

^{13}C NMR (101 MHz, DMSO-d_6) δ 139.44, 134.00, 129.99, 127.53, 115.56, 112.76, 45.33, 39.56, 39.36, 36.59, 29.06, 28.28, 26.92, 26.53, 26.38, 20.79, 20.74 ppm.

^{31}P NMR (202 MHz, DMSO-d6) δ 26.42 ppm.

HRMS (m/z): $[M+H]^+$ calcd. for$C_{19}H_{23}Br_2NO_3P$ 501.9777, found: 501.9779.

Elemental analysis: calcd. for $C_{19}H_{22}Br_2NO_3P$, %: C 45.35, H 4.41, N 2.78, found, %: C 45.65, H 4.47, N 2.66.

Figure S2 DFT calculation of steric effect

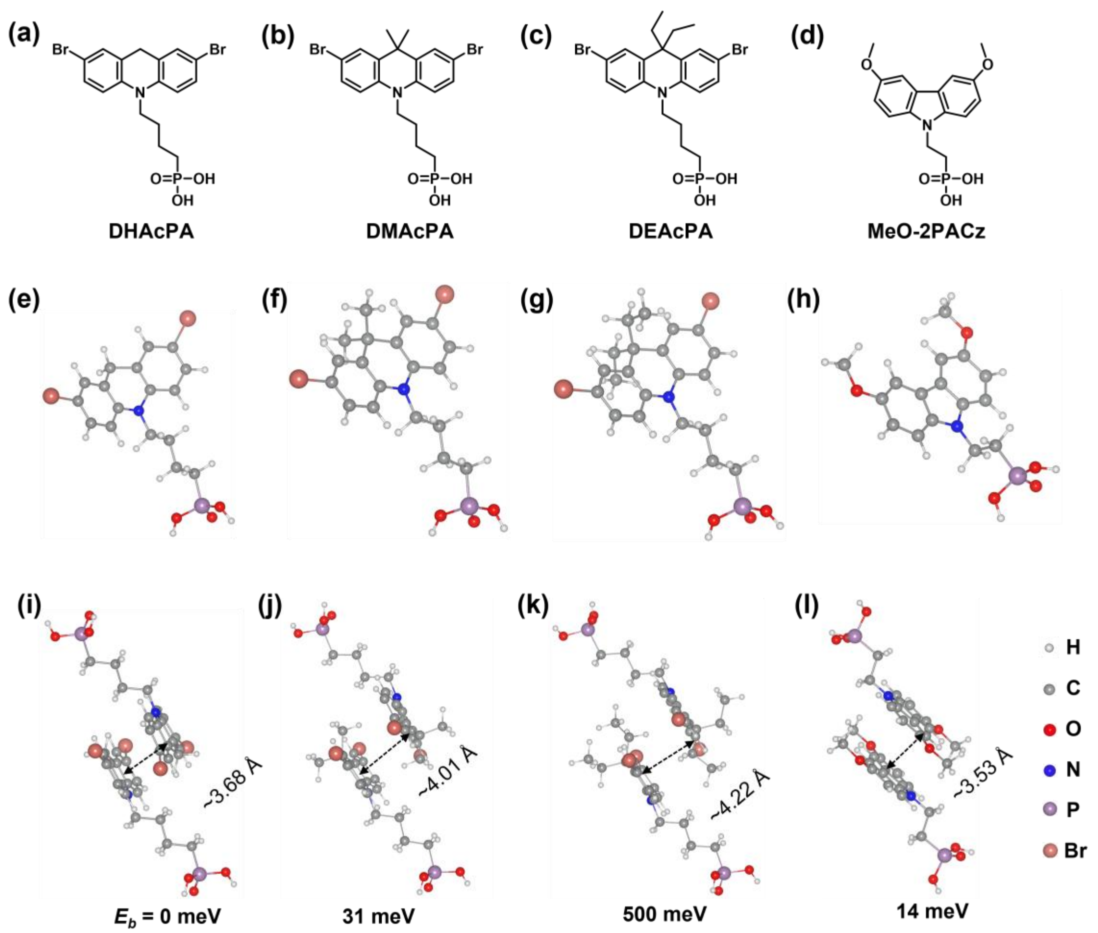


Figure S2. DFT calculation of steric effect. (a) (b) (c) (d) are the chemical structures, (e) (f) (g) (h) are the optimized molecular structure, and (i) (j) (k) (l) are the corresponding dimers of DHAcPA, DMAcPA, DEAcPA and MeO-2PACz.

That steric effect can be predicted by Density Functional Theory (DFT) calculation on π-π stacking of molecules. Basing on a dimer model, AcPA molecules with different branch chain sizes from dihydrogen (DHAcPA), dimethyl (DMAcPA) to diethyl (DEAcPA) were screened. DHAcPA show the lowest binding energy and easily to stack and aggregate. The DHAcPA has the smallest layer distance of ~3.68 Å and the lowest binding energy (set as 0 meV for comparison) of three materials, which indicates that DHAcPA is easily aggregate in the perovskite film deposition process. Moreover, the aggregate organic molecules would be a potential defect in the perovskite film and become a recombination center, and finally result in a bad photovoltaic performance and poor device stability. With dimethyl group (Fig. S2 j), the binding energy of DMAcPA dimer is 31 meV higher than DHAcPA, larger than the thermal fluctuation energy (~26 meV). Its distance ~4.07Å is also far from 3.35 Å of typical π-π stacking. For DEAcPA, the layer distance is ~4.22 Å and the binding energy of dimer is 500 meV. The largest layer distance and binding energy indicate that DEAcPA is difficult to form intermolecular interaction and probably suppress charge transport process. Besides, the DEAc is not a commercial chemical product and difficult to synthesis. Compared to

DHAcPA and DEAcPA, DMAcPA with layer distance of ~4.01 Å and binding energy of 31 meV, the appropriate steric effect would be beneficial for DMAcPA dispersion in perovskite precursor solution and restrain the molecular aggregation. The excellent dispersibility of DMAcPA with relatively lower dissolution rate would finally distribute on the grain boundary and is brought to the bottom surface of perovskite in the anti-solvent assisted film deposition process.

We also calculate the MeO-2PACz as a benchmark, which is a commercial material with phosphonic acid group and widely used in perovskite solar cells. As shown in figure S11, MeO-2PACz dimer with the smallest layer distance of ~3.53 Å and a small binding energy of 14 meV, which indicate that MeO-2PACz has strong intermolecular interaction and is easy to aggregate in perovskite precursor solution.

NMR spectra

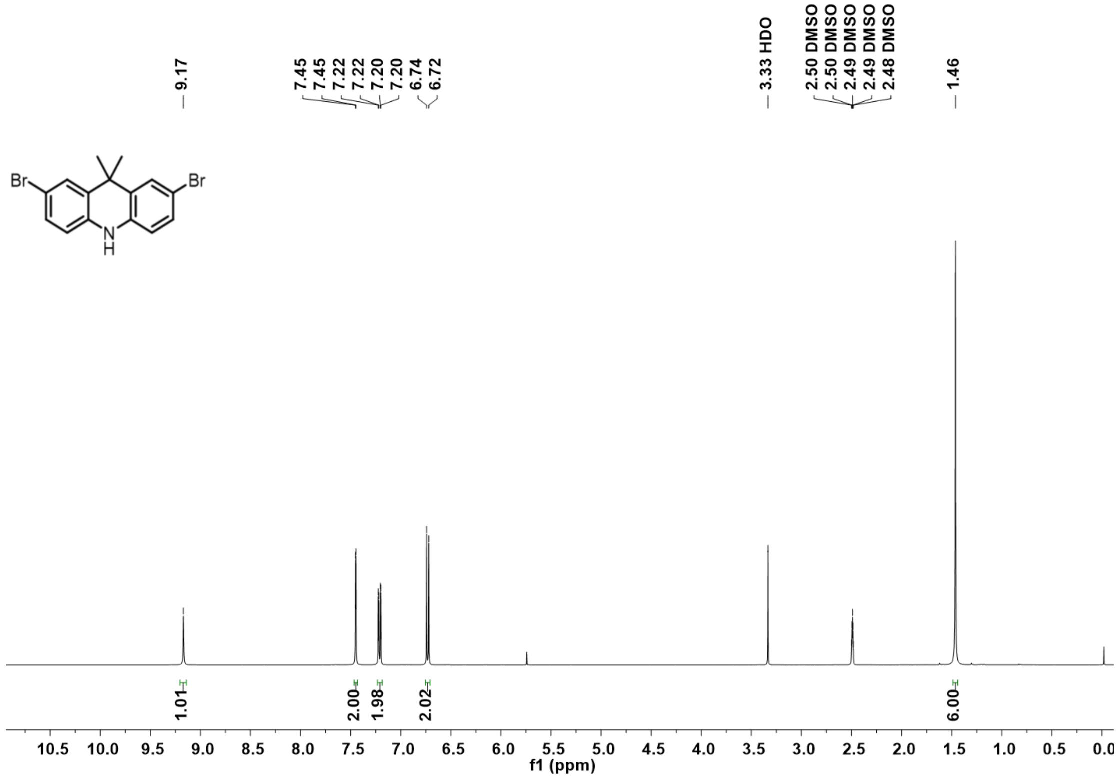


Figure S3. ^{1}H NMR spectrum of intermediate **2**.

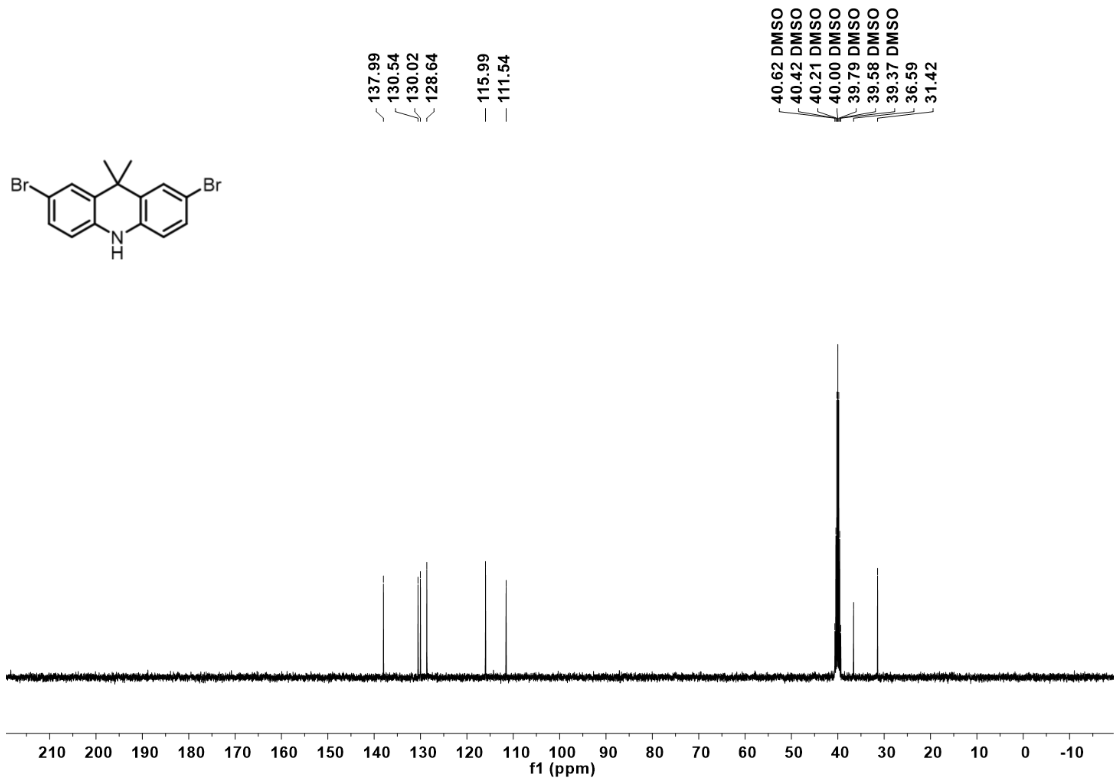


Figure S4. ^{13}C NMR spectrum of intermediate **2**.

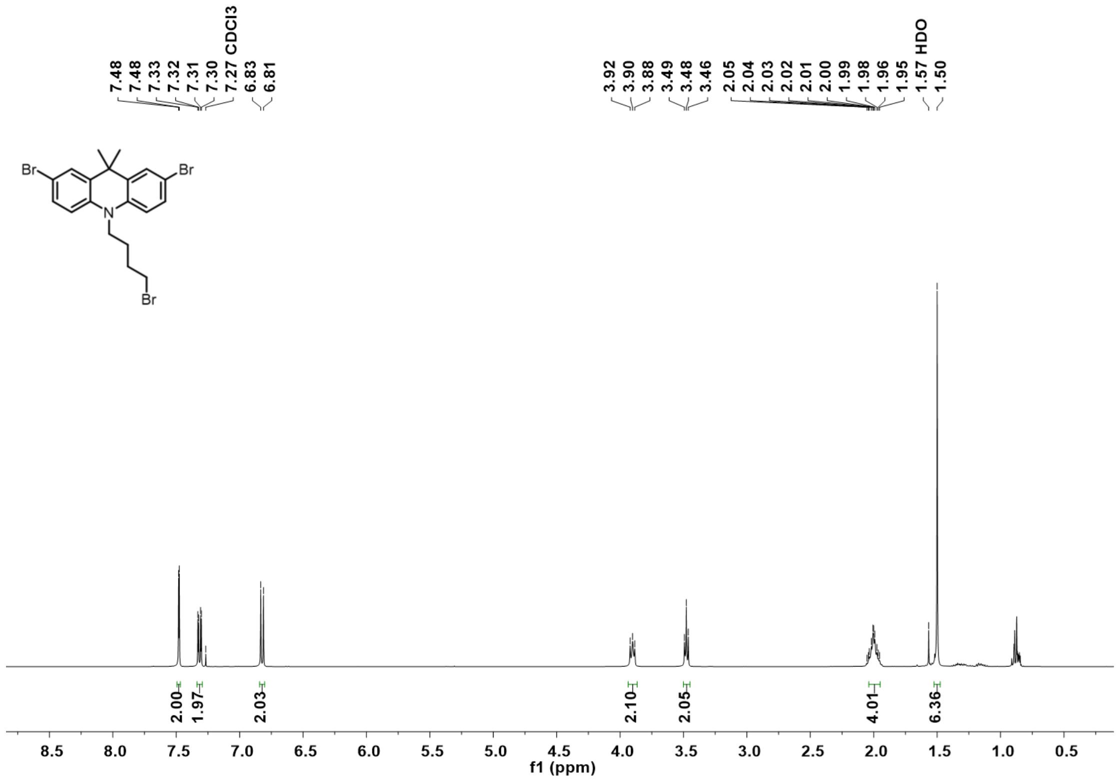


Figure S5. ^{1}H NMR spectrum of intermediate **3**.

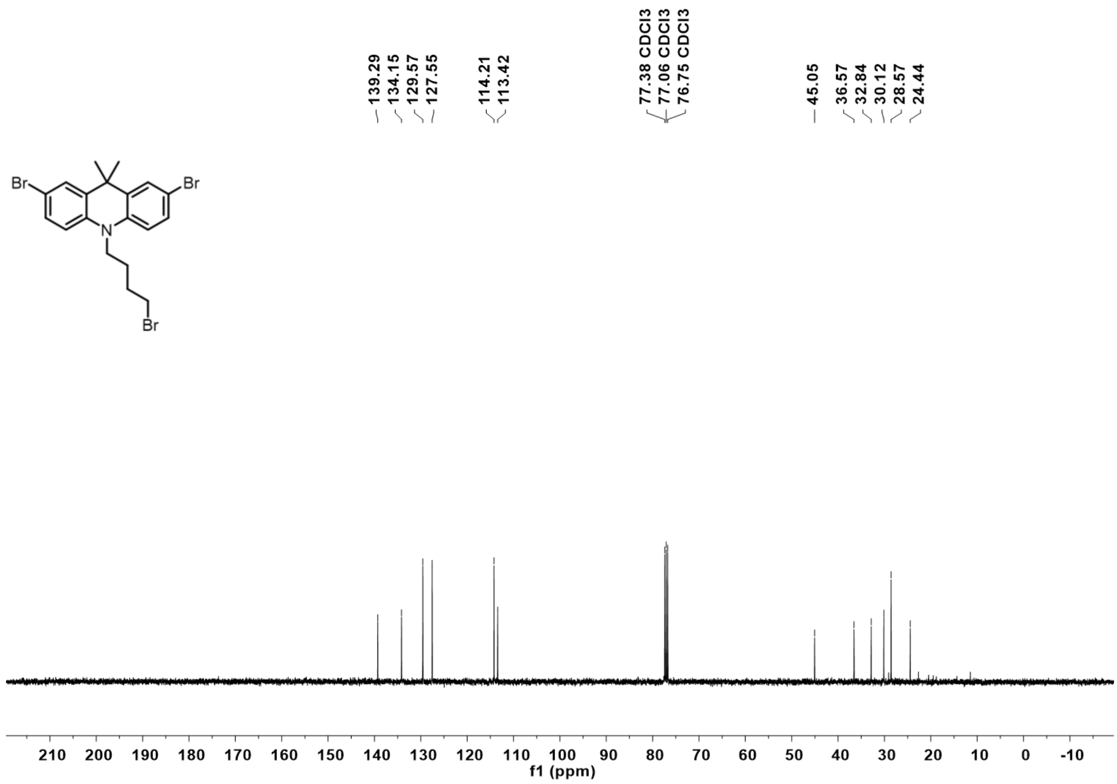


Figure S6. ^{13}C NMR spectrum of intermediate **3**.

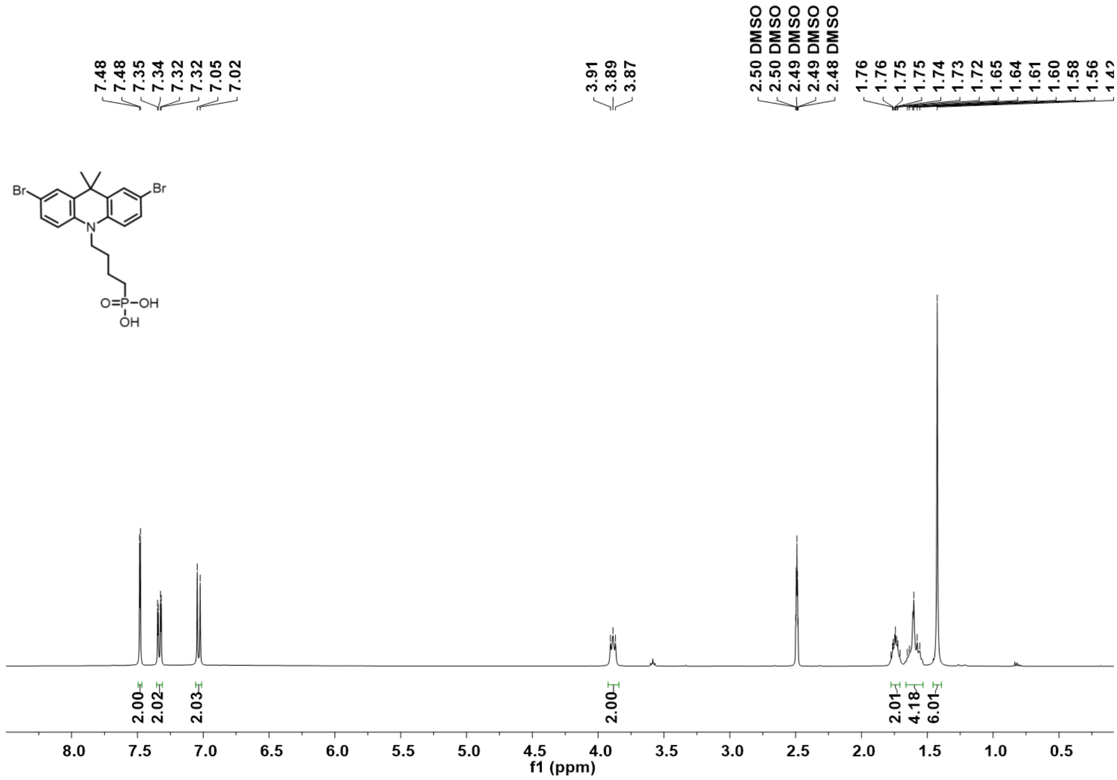


Figure S7. ^{1}H NMR spectrum of DMAcPA.

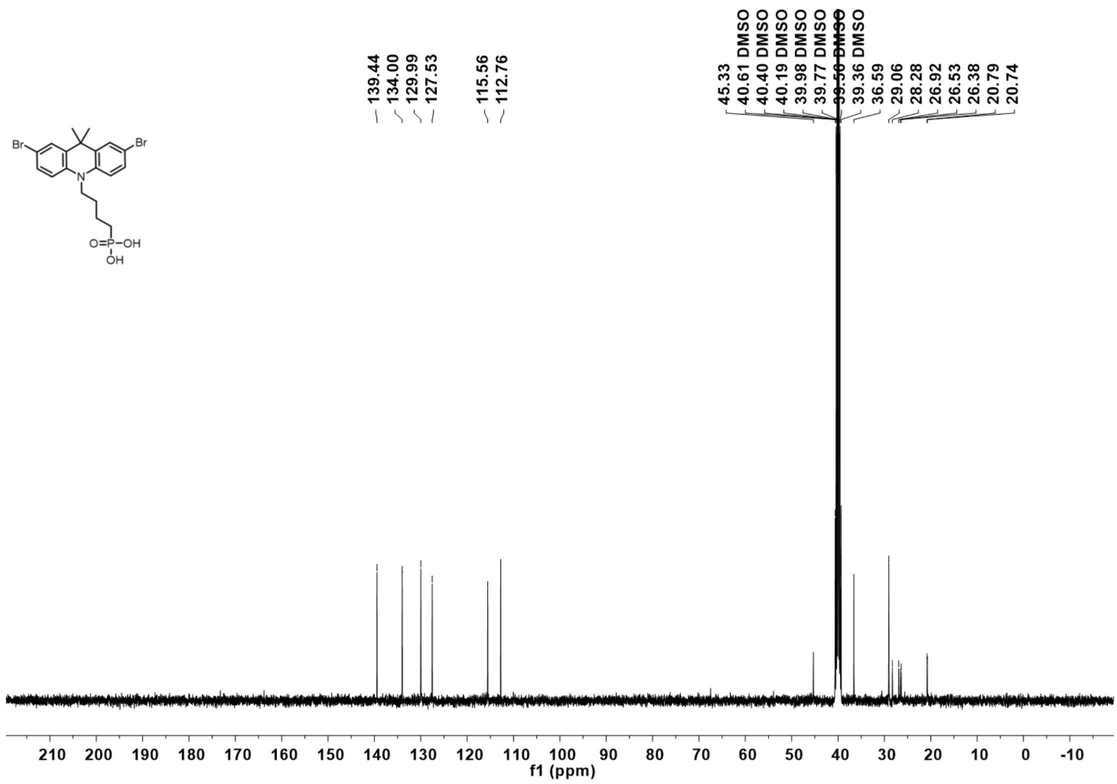


Figure S8. ^{13}C NMR spectrum of DMAcPA.

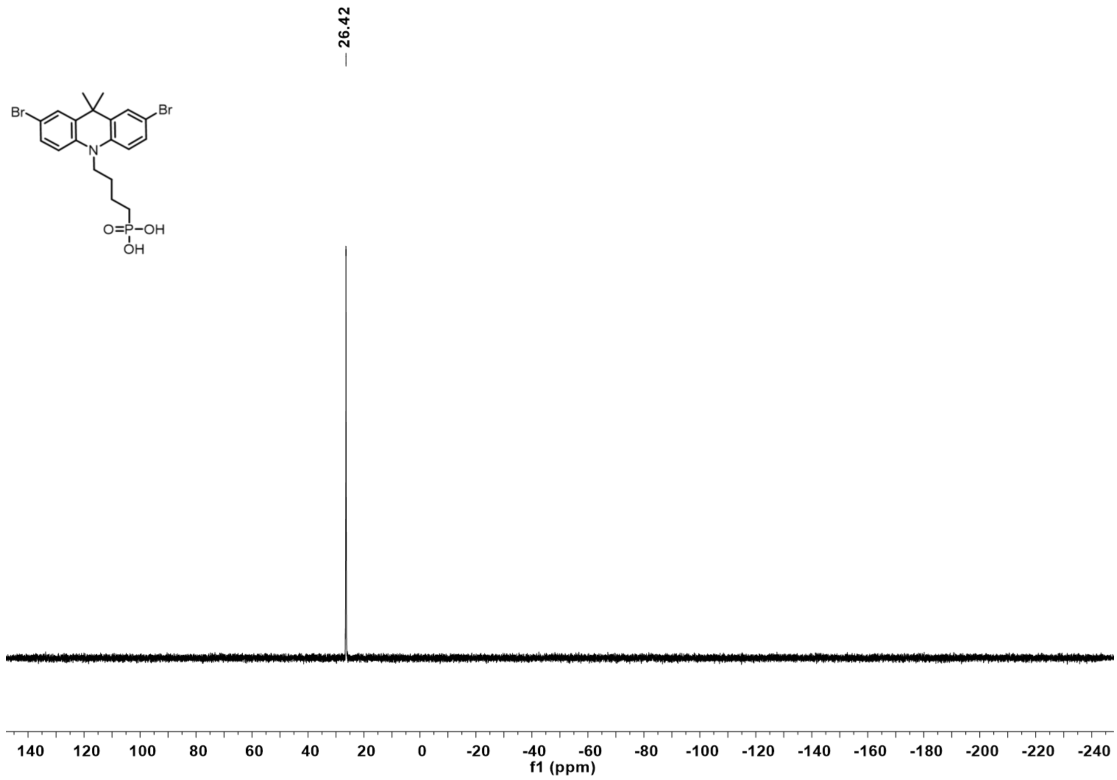


Figure S9. ^{31}P NMR spectrum of DMAcPA.

Figure S10 TGA of DMAcPA

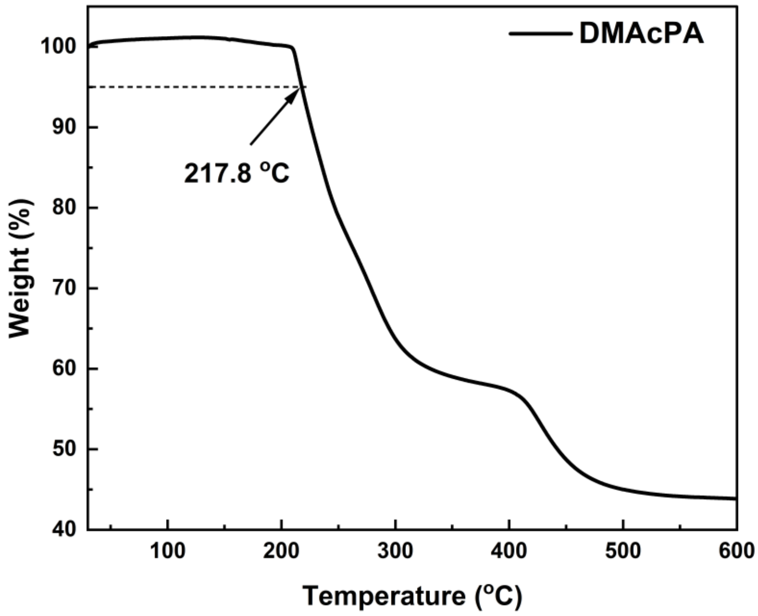


Figure S10. Thermal gravimetric analysis (TGA) curve of DMAcPA with a heating rate of 10 ℃/min.The thermal decomposition temperature (T_d, 5% weight loss) of DMAcPA at 217.8 ℃ represents a good thermal stability.

Figure S11 UV-vis absorption spectra of DMAcPA.

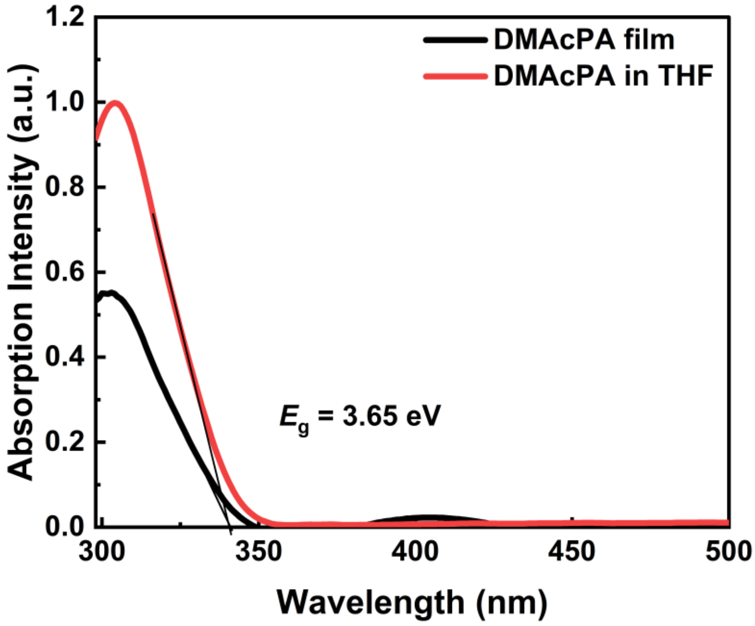


Figure S11. UV-vis spectra of DMAcPA in THF solution (10^{-5} M) and films. The optical bang gap (E_g=3.65 eV) of DMAcPA was calculated from the equation of E_g = $1240/\lambda_{onset}$ (eV), where theonset wavelength of absorption spectra is 340 nm.

Figure S12 DFT calculation of HOMO and LUMO

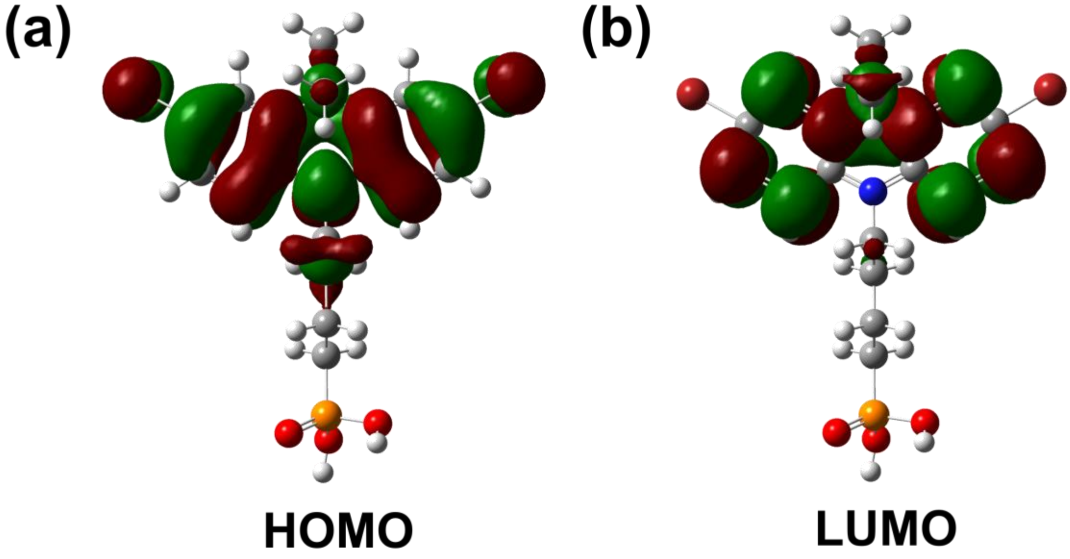


Figure S12. DFT calculation of DMAcPA based on B3LYP/6-311G*. Electron cloud distributions of (a) HOMO and (b) LUMO.

Figure S13 UPS of DMAcPA

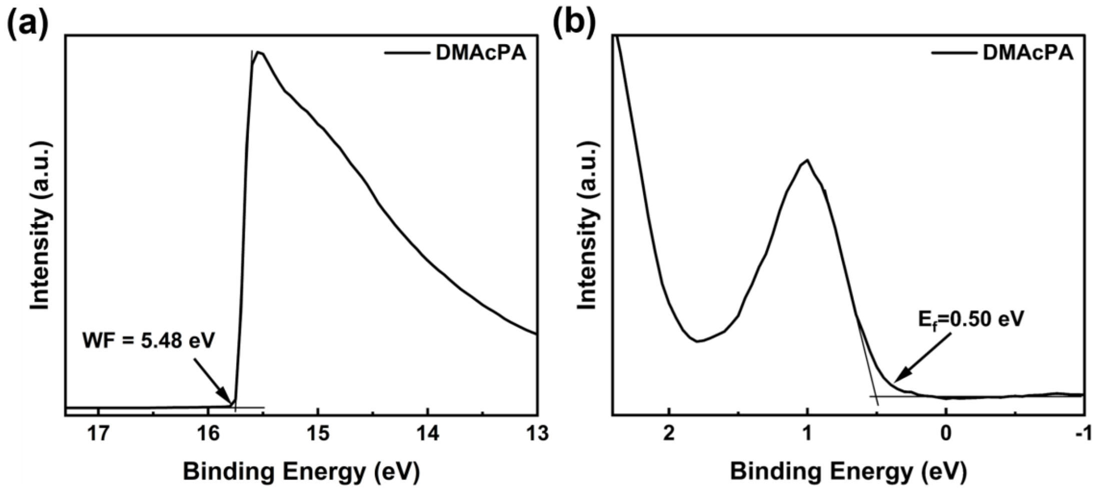


Figure S13. The (a) cut-off region and (b) onset region of ultraviolet photoelectron spectroscopy (UPS) of DMAcPA films. The work function (WF) of DMAcPA was calculated by the equation of WF=21.22 eV-$E_{cut\text{-}off}$ = 5.48 eV. The HOMO of DMAcPA was calculated by the equation of HOMO=-[21.22 eV-($E_{cut\text{-}off}$-E_f)] = -5.98 eV. By combining the optical band gap (E_g=3.65 eV) of DMAcPA in Figure S10, the LUMO of DMAcPA was calculated as -2.33 eV by the equation of LUMO=HOMO+Eg. DMAcPA with a high LUMO energy level would effectively suppress the electron transporting to the ITO electrode and decrease the recombination energy loss at the interface.

Figure S14 UV-vis absorption spectra and tauc plots of the three kinds of perovskite films

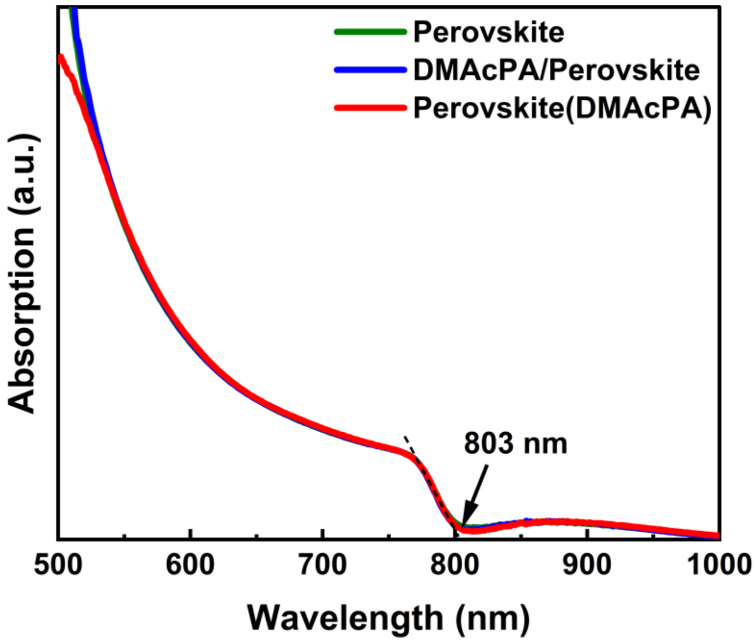


Figure S14. UV-vis spectra of the three kinds of perovskite films deposited on ITO substrates. The Perovskite means the pristine undoped perovskite film directly deposited on ITO substrate. The DMAcPA/Perovskite means the pristine undoped perovskite film deposited on a DMAcPA layer coated ITO substrate. The Perovskite(DMAcPA) means the DMAcPA doped perovskite film directly deposited on ITO substrate. All the three kinds of perovskite films exhibit the same absorption edge of 803 nm in wavelength, which indicates that large-size DMAcPA can't enter the lattice and change the phase.

Figure S15 PL spectra of the three kinds of perovskite films

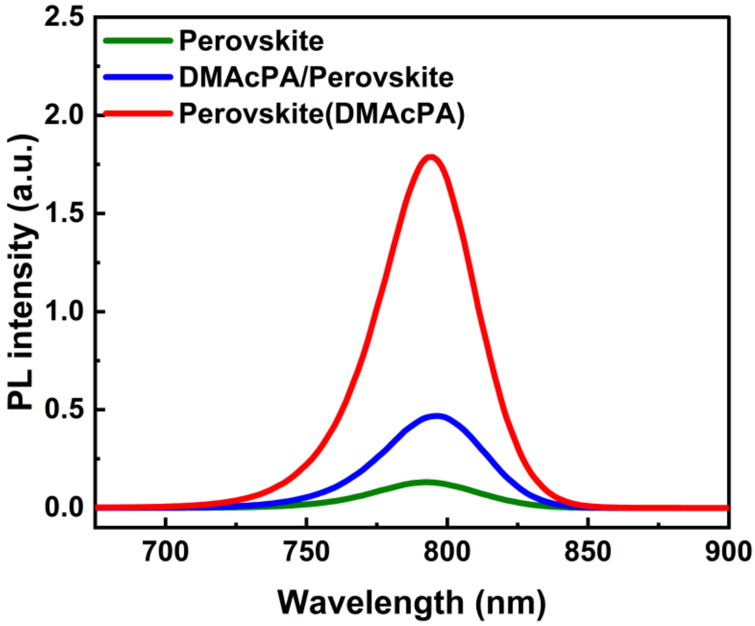


Figure S15. Steady state photoluminescence (PL) spectra of the three kinds of perovskite films deposited on glass.Compared with thePerovskite sample, the PL intensity of DMAcPA/Perovskite and Perovskite(DMAcPA) with a little and huge enhancement, respectively, represented higher crystallinity and lower trap state density. The steady state PL result was in good agreement with the XRD and SEM results in the main body of manuscript.

Table S1 Fitting results of the TRPL spectra of Perovskite, DMAcPA/Perovskite and Perovskite(DMAcPA) films deposited on glass substrate.

	A_1	τ_1 (ns)	A_2	τ_2 (ns)	τ_{avge} (ns)
Perovskite	743.11	14.88	387.77	524.56	498.28
DMAcPA/Perovskite	147.39	31.42	1295.12	949.80	946.38
Perovskite(DMAcPA)	204.96	5.04	1115.19	1927.75	1926.83

The TRPL decay spectra were fitted by a biexponential function:

$$y = y_0 + A_1 e^{-x/\tau 1} + A_2 e^{-x/\tau 2}$$

and the average decay lifetime was calculated with function:

$$\tau_{average} = \frac{A_1\tau_1^2 + A_2\tau_2^2}{A_1\tau_1 + A_1\tau_2}$$

Figure S16 XRD patterns of PbI_2 films with different dosage of DMAcPA

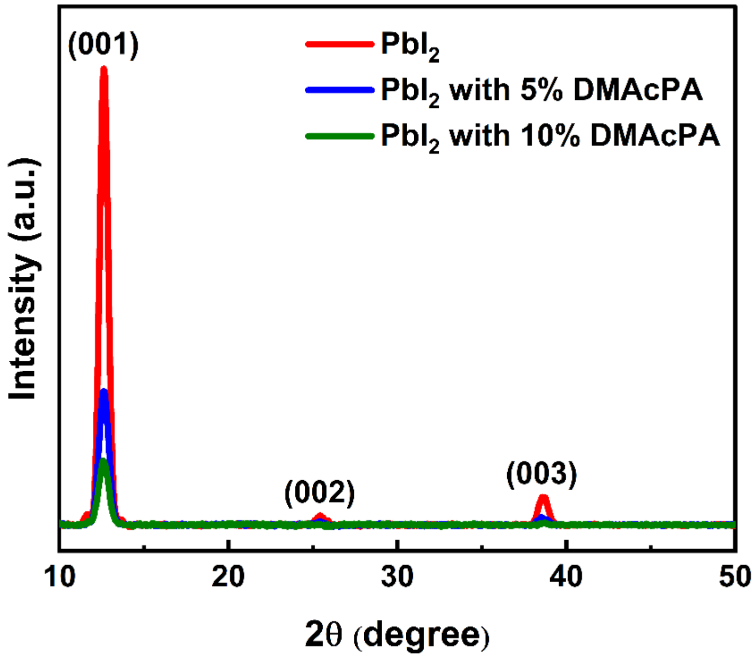


Figure S16. The XRD patterns of pristine PbI_2 film and doped PbI_2 film with different molar ratio DMAcPA. As shown, the peak intensity declined significantly when the films were doped by DMAcPA.

Figure S17 Peel-off method for the buried bottom surfaces of Perovskite films

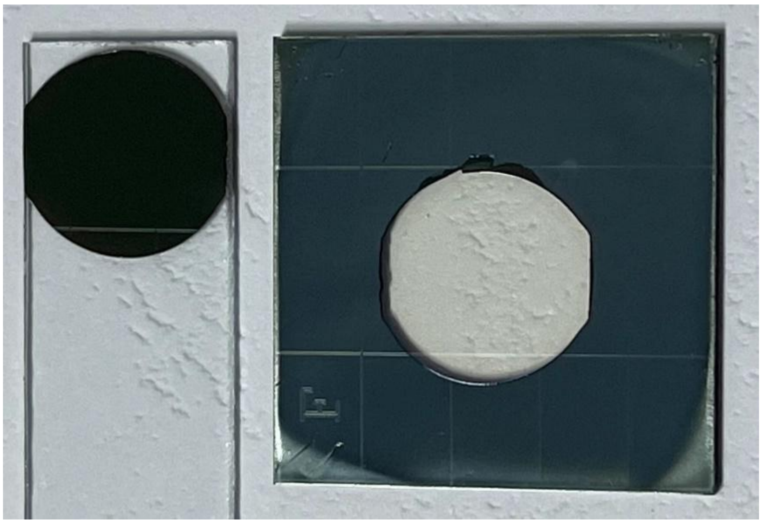


Figure S17. The photography of perovskite film for bottom surface characterization. The bottom surface sample of perovskite film was fabricated with the following method: the top surface of prepared perovskite film was partly pasted together with a glass slide through UV glue. After the glue is solidified, the film was peeled off from the ITO substrate, then the bottom surface of perovskite film for characterization was got.

Figure S18 EDX characterization between PbI_2 flakes and common grains of prinstine perovskite films

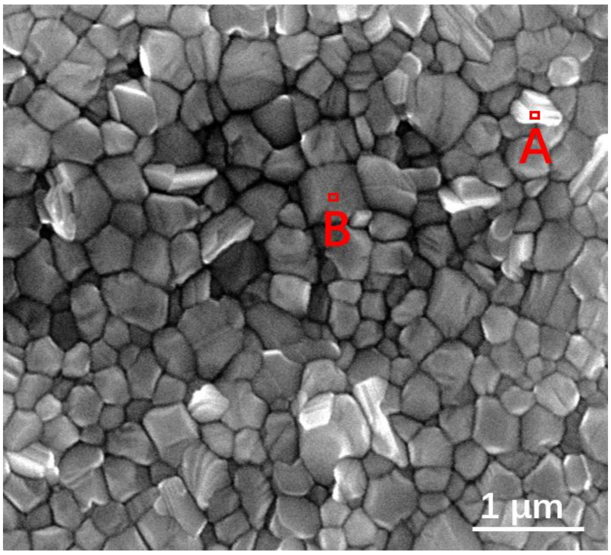


Figure S18. Energy dispersive X-ray spectroscopy (EDX) collected from the bright (PbI_2 flakes, A) and dark region (common grain, B) in the SEM of pristine perovskite film. The element analysis results are summarized in table S2 and S3, respectively.

Table S2 Data of EDX of position A

Element	Weight percentage	Atomic percentage
C K	4.81	39.16
Pb M	41.87	19.76
I L	53.32	41.08
Total	100.00	

Table S3 Data of EDX of position B

Element	Weight percentage	Atomic percentage
C K	6.43	45.93
Pb M	35.15	14.56
I L	58.42	39.51
Total	100.00	

Figure S19 Contact angle of substrate surfaces

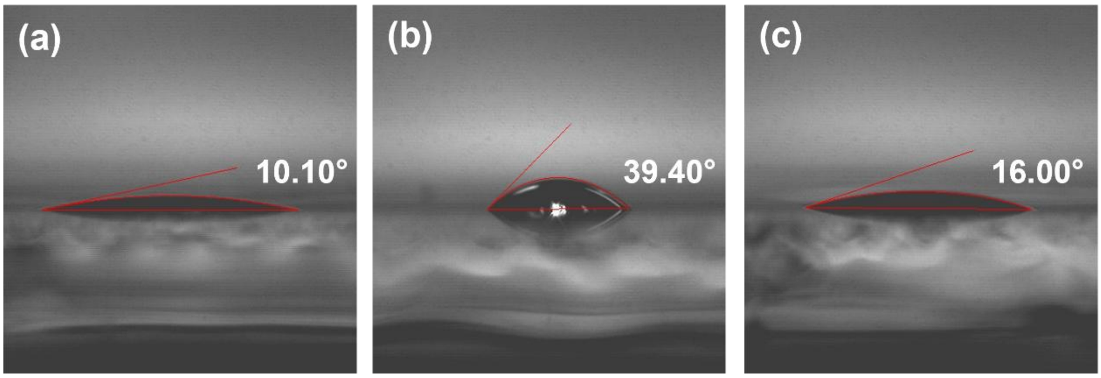


Figure S19. Contact angles of perovskite precursor solution on (a) ITO and (b) ITO/DMAcPA substrates, and (c) perovskite precursor solution with DMAcPA on ITO substrate. The contact angle of perovskite precursor solution on ITO and ITO with DMAcPA are 10.10 °and 39.40 °, respectively, indicating a poor wettability of the latter. However, the perovskite precursor solution doped with DMAcPA has a contact angle of 16.00 °, which is close to the bare ITO substrate with pure perovskite precursor solution. The above results demonstrate that the DMAcPA doped perovskite precursor solution has the suitable wetting ability with bare ITO to grow compact perovskite crystalline films.

Figure S20 FTIR spectra of DMAcPA and pristine undoped perovskite films

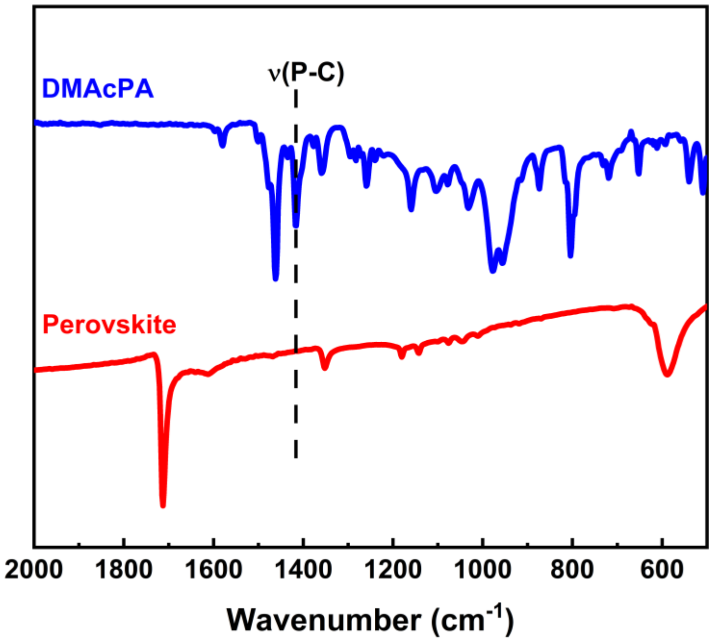


Figure S20. FTIR spectra of pristine undoped perovskite and DMAcPA films. In order to investigate the distribution of DMAcPA at interfaces, AFM-IR was employed in this work. A feature peak of P-C stretching vibration of DMAcPA at 1415 cm-1 was selected for AFM-IR measurements to exclusively display DMAcPA on the perovskite film surfaces.

Figure S21 XPS of two kinds of detached ITO surfaces.

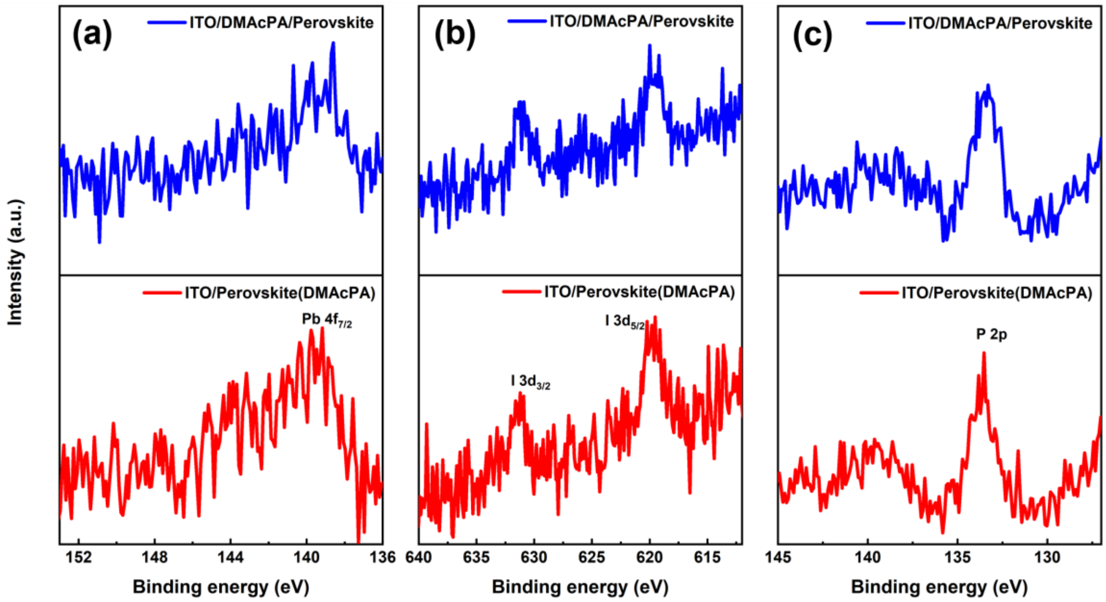


Figure S21. XPS spectra of Pb 4f (a), I 3d (b) and P 2p (c) at the detached ITO surface from ITO/DMAcPA/Perovskite (blue) and ITO/Perovskite(DMAcPA) (red) samples.

Figure S22 SEM images of the DMAcPA doped perovskite film with the dosage of 10mg/ml

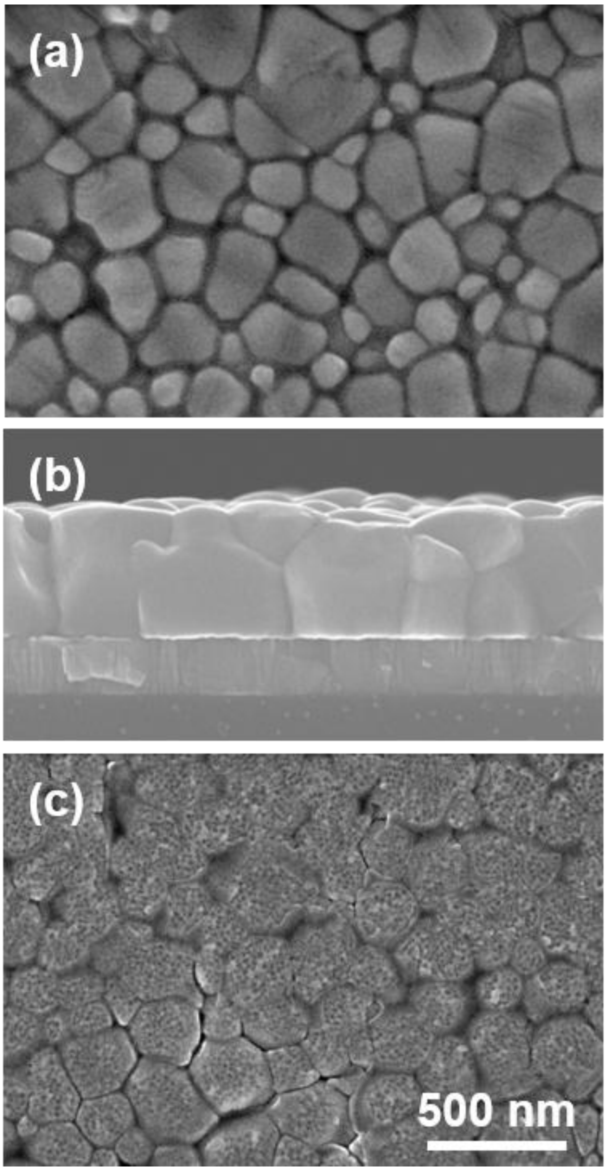


Figure S22. SEM images of the doped perovskite film with a dosage of 10 mg/mL DMAcPA: (a) top surface, (b) cross section and (c) buried bottom surface. Obviously, the grain boundaries are expanded with impressive gaps in both the top (a) and bottom surfaces (b). The grain surfaces at the bottom become much rougher further prove an autogenic layer of DMAcPA and PbI_X coordinated agglomerates extruded to the bottom surface owing to the molecule-extrusion process.

Table S4 pHtest results of DMF/DMSO (4:1) mixed solvent, FAI, PbI_2 and Perovskitecontained solutions with the identical composition and concentration, without (Left) and with 10 mg DMAcPA molecules (Right) dissolved in.

Solution Kind	pH	Solution Kind	pH
DMF/DMSO	**8.40**	**+DMAcPA**	**6.25**
FAI	**7.80**	**FAI+DMAcPA**	**4.30**
PbI_2	**9.44**	**PbI_2+ DMAcPA**	**3.34**
Perovskite	**7.57**	**Perovskite+ DMAcPA**	**4.12**

pH test is a good method to reveal the concentration of solvated protons[20]. Table S4 shows the pH test results of DMF/DMSO (4:1) mixed solvent, FAI, PbI_2 and PVK contained solutions with the identical composition and concentration to our process. The data show clearly the dissolution of DMAcPA molecules results in the increase of solvated $[H^+]$ in each solution, which ambiguously reflects the deprotonating of DMAcPA happens in its dissolving process. Both PbI_2 and FAI can stimulate the ionization degree of DMAcPA. In the PVK solution, the solvated $[H^+]$ can increase by over 3 orders after dissolving DMAcPA, with the pH value rising from 7.57 to 4.12.

Figure S23 PbI_2 concentration dependent shift of active hydrogen of DMAcPA in DMF-d7

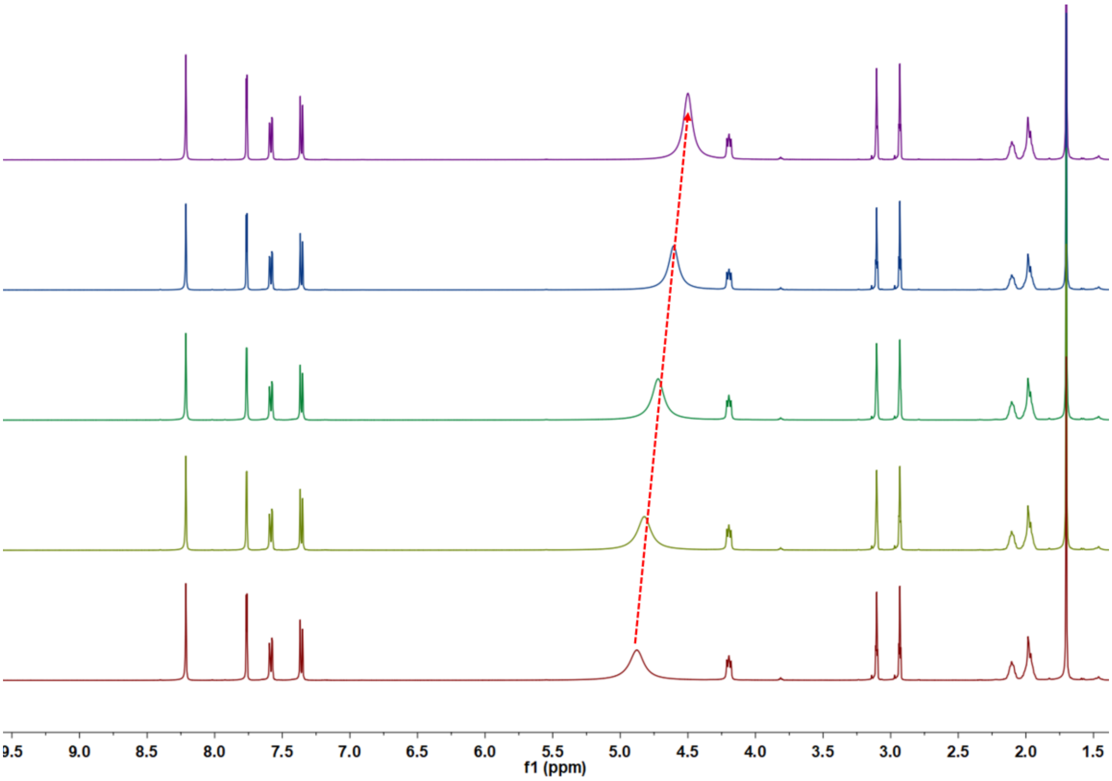


Figure 23. PbI_2 concentration dependent shift of active hydrogen of DMAcPA in DMF-d7. PbI_2 dosage increase sucessively from bottom to top, and the active hydrogen peak moves consectively to low chemical shift or highfield.

Larger shift caused by PVK can be attributed to higher concentration of iodine ions input, which is in agreement with the enhanced ionization of DMAcPA by iodine ions in the above pH tests.

Figure S24^{13}P NMR results of DMAcPA invovled solutions

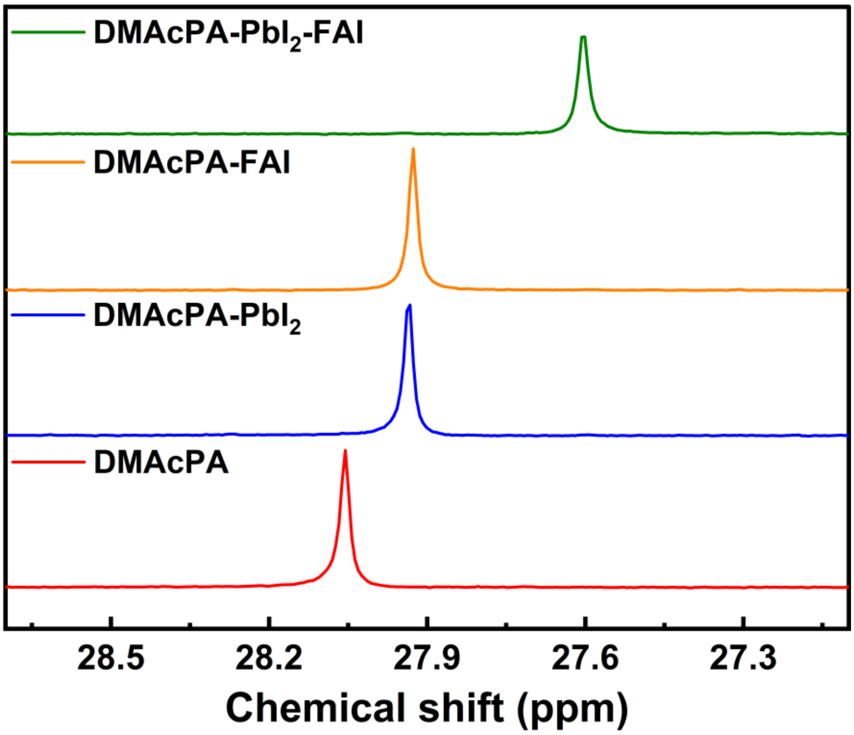


Figure S24. ^{31}P NMR results of DMAcPA invovled solutions. DMAcPA-only in DMF-d7 (red); DMAcPA+PbI_2 in DMF-d7 (blue); DMAcPA+FAI in DMF-d7 (yellow); DMAcPA+FAI+PbI_2 in DMF-d7 (green). The up-field shift of ^{31}P resonance signal from 28.06 ppm to 27.93 ppm in DMF solution with dissolving those iodides, demonstrates the deprotonated PA group is probably influenced by the H--I hydrogen bond.

Figure S25 DFT calculation of neutral DMAcPA absorbed on perovskite surface

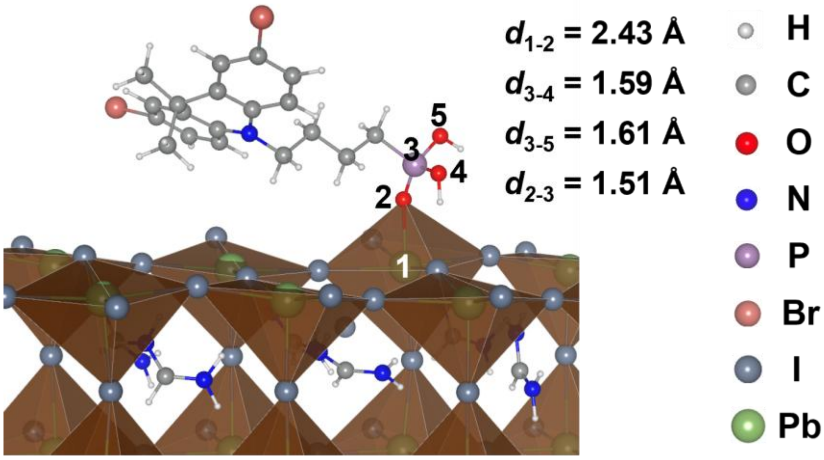


Figure S25. DFT calculation of neutral DMAcPA absorbed on perovskite surface. Stable adsorption structure on the PbI_2-terminated $FAPbI_3$ surface by the P=O--Pb dative bond.

Since the negative charge of the adsorbed deprotonated DMAcPA must be balanced in real experiments, we add a H^+ near the adsorption site to mimic the spatial effect of solvated cations, e.g. $H^+[DMF]_x$ or $H^+[DMSO]_x$, which prevents a direct contact with the deprotonated DMAcPA.

Table S5 The bond length variation of original DMAcPA and the ionized DMAcPA absorbed on perovskite crystal surface (Fig. 3B) after optimization.

Neutral DMAcPA		Ionized DMAcPA absorbed on perovskite	
Bond type	Bond length (Å)	Bond type	Bond length (Å)
P-O (2-3)	1.63	Pb-O (1-2)	2.18
P-O (3-4)	1.62	P-O (2-3)	1.57
P=O (3-5)	1.49	P-O (3-4)	1.64
		P=O (3-5)	1.49
		I-H (6-7)	1.88
		I-H (7-8)	1.98

Figure S26 XPS of buried bottom surfaces of two kinds of perovskite films

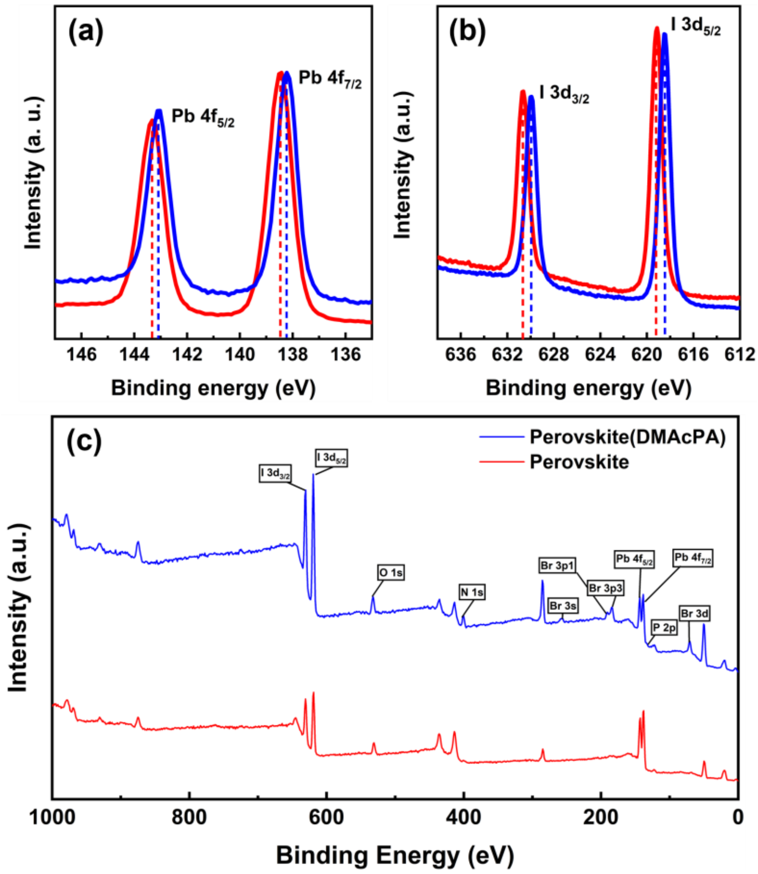
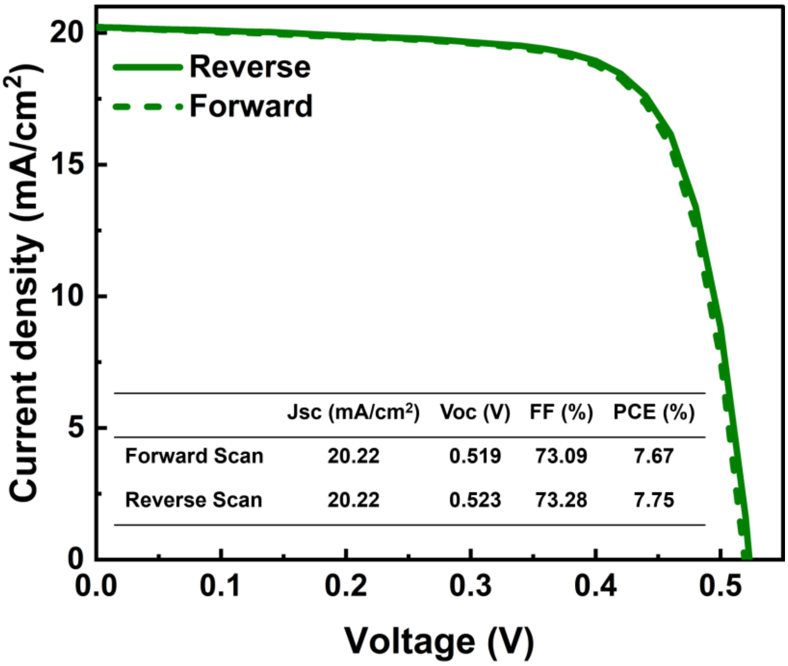


Figure S26. XPS spectra of Pb 4f (a), I 3d (b) and survey (c), detected at the bottom surfaces of pristine (red) and DMAcPA dopped (blue)perovskite films, identical to the fabrication process reported in the main text. The red shift in binding energy of Pb[25] detected at the buried bottom surface of perovskite (Fig. S26a) can also indicate the O--Pb dative bond weakens the binding energy of the main-flow Pb-I covalent bond and herein accounts for the red shift of Pb. Regarding the I shift (Fig. S26b), it can be attributed to hydrogen bonding of P-O-H--I, which has been well discussed and examined by the above down-field chemical shift of ^{1}H NMR signal (Fig. 3A).

Figure S27 Photovoltaic performance of the device based on the ITO/Perovskite



	Jsc (mA/cm²)	Voc (V)	FF (%)	PCE (%)
Forward Scan	20.22	0.519	73.09	7.67
Reverse Scan	20.22	0.523	73.28	7.75

Figure S27. J-V curves and photovoltaic performance parameters of the champion PSC with ITO/Perovskite/$PC_{61}BM$/BCP/Ag structure. A champion PCE of 7.75% was achieved when reverse scan, with a Jsc of 20.22 mA/cm^2, a Voc of 0.523 V and an FF of 73.28%.

Figure S28 Optimization of the concentration of DMAcPA modified ITO

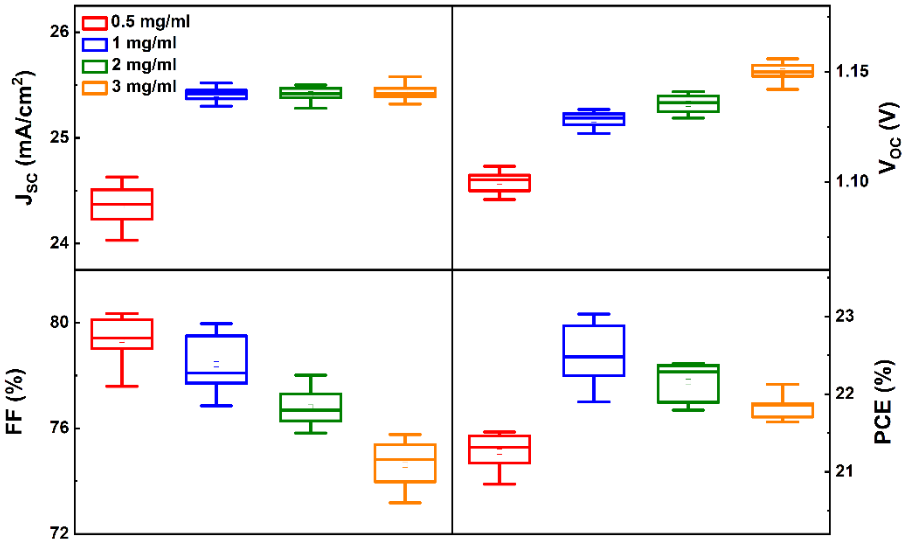


Figure S28. Photovoltaic performance of PSCs with different concentration of DMAcPA modified ITO substrate. The centre line represents the median value, the small circle indicates the mean value, the box gives the standard deviation, and the whiskers represent the minimum and maximum values. In the statistics, 24 devices were used for each kind of devices. The thickness of DMAcPA film coating on ITO substrate was optimized by turning the concentration of DMAcPA in dry THF solution. With the increasement of DMAcPA from 0.5 mg/mL to 3 mg/mL, the V_{OC} was gradually increased while the FF was decreased. The J_{SC} of devices represented a significant enhancement when the concentration of DMAcPA increased from 0.5 mg/mL to 1 mg/mL, then keep the same value. Above all, the champion PCE (23.03%, average PCE of 22.50%) of the DMAcPA/Perovskite device was achieved with optimal 1 mg/mL DMAcPA.

Figure S29 Steady Power Output (SPO) of the target and the control PSCs

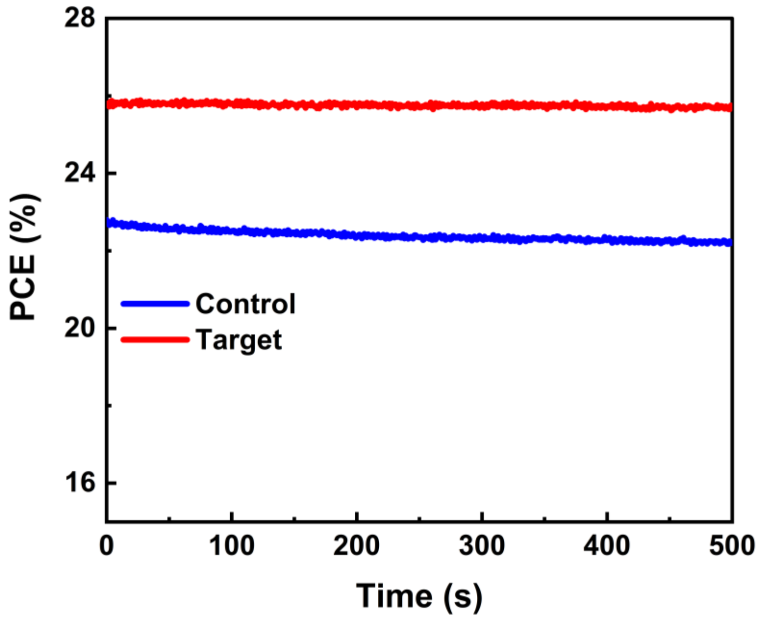


Figure S29. Steady power output (SPO)testedfor at a duration of 500 s at 0.96 eV and 1.04 V for the DMAcPA/Perovskite (Control) and the Perovskite(DMAcPA) (Target)based champion devices, respectively.This results confirms the reliability of our J−V measurements.

Figure S30 Statistic device parameters of the target and the control PSCs

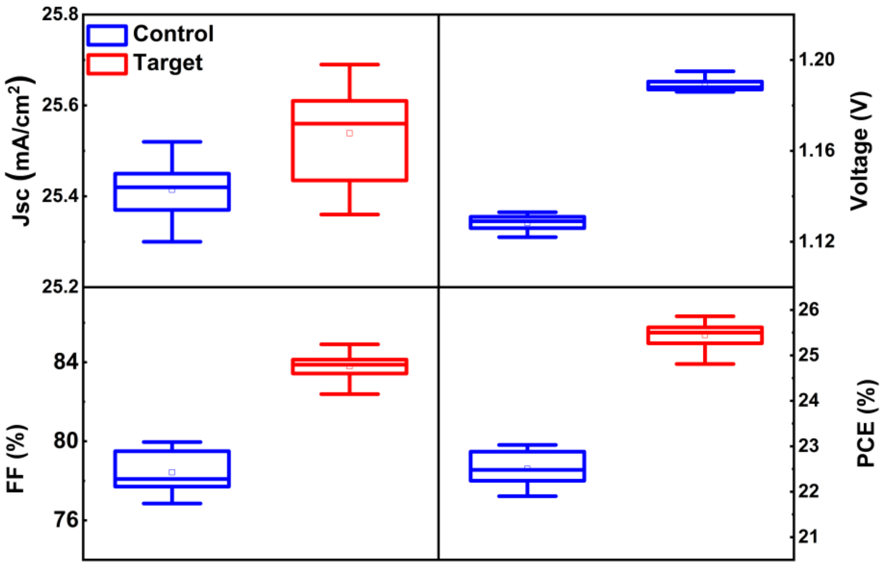


Figure S30. Statistic photovoltaic parameters of PSCs based on DMAcPA/Perovskite (Control) and Perovskite(DMAcPA) (Target) in each batch of 24 devices. The center line represents the median value, the box gives the standard deviation, and the whiskers represent the minimum and maximum values).

Table S6 Corresponding numbers of the above statistic plots of the Control and Target PSCs. Parameters of the champion devices were listed in the parentheses.

	J_{SC} (mA/cm^2)	V_{OC} (V)	FF (%)	PCE (%)
Control	25.41±0.07 (25.45)	1.128±0.004 (1.133)	78.43±1.11 (79.87)	22.50±0.38 (23.03)
Target	25.54±0.10 (25.69)	1.189±0.003 (1.194)	83.79±0.68 (84.91)	25.45±0.28 (25.86)

The photovoltaic performance parameters of Control and Target devices have a narrow distribution, which demonstratesgood reproducibility of our fabrication process.

Figure S31 Certification report

Test and Calibration Center of New Energy Device and Module,
Shanghai Institute of Microsystem and Information Technology,
Chinese Academy of Sciences (SIMIT)
235, Chengbei Road, Jiading, Shanghai, China

CMA 170017163974 · ilac-MRA · CNAS TESTING CNAS L8490

Measurement Report

Client Name	Southern University of Science and Technology (SUSTech)
Client Address	1088 Xueyuan Avenue, Shenzhen 518055, China
Sample	Inverted Perovskite Solar Cell
Manufacturer	Shenzhen Key Laboratory of Full Spectral Solar Electricity Generation (FSSEG)
Application	SIMITL72022112401
Measurement Date	24th November, 2022

Performed by:	Qiang Shi	Date:	24/11/2022
Reviewed by:	Wenjie Zhao	Date:	24/11/2022
Approved by:	Zhengxin Liu	Date:	25/11/2022

The measurement report without signature and seal are not valid. This report shall not be reproduced, except in full, without the approval of SIMIT.

Report No.22TR112401 1/4

Sample Information

Sample Type	Inverted Perovskite Solar Cell
Serial No.	1#
Lab Internal No.	22112401-1#
Measurement Item	I-V characteristic
Measurement Environment	21.9±2.0°C, 36.5±5.0%R.H

Measurement of I-V characteristic

Reference cell	AK-200
Reference cell Type	mono-Si, WPVS, calibrated by National Institute of Metrology, China (Certificate No. GXgf2022-01035)
Calibration Value/Date of Calibration for Reference cell	128.1mA/ Apr. 2022
Measurement Conditions	STC, linear sweep based on IEC 60904-1:2006
Measurement Equipment/ Date of Calibration	Steady State Solar Simulator (YSS-T155-2M) / July.2022 IV test system (ADCMT 6246) / June. 2022 SR Measurement system (CEP-25ML-CAS) / April.2022
Mismatch Factor	SMM=0.9897

Serial Number	Scan Mode	Area*(da) (cm^2)	Isc (mA)	Voc (V)	Pmax (mW)	FF (%)	Eff (%)
1#	Isc to Voc	0.0802	2.056	1.186	2.032	83.34	25.33
	Voc to Isc	0.0802	2.058	1.189	2.036	83.24	25.39

Supplementary information: *(da), Designated area defined by thin metal aperture mask.

Test results listed in this measurement report refer exclusively to the mentioned test sample.
The results apply only at the time of the test, and do not imply future performance.

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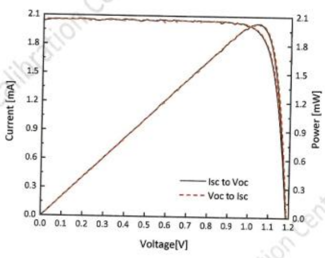
======== I-V Curve ========



Date:	24th November,2022	Date:	24th November,2022
Data No:	IV_22112401-1#_D	Data No:	IV_22112401-1#_R
Serial No:	1#	Serial No:	1#
Area[da]	0.0802 cm^2	Area[da]	0.0802 cm^2
Isc	2.056 mA	Isc	2.058 mA
Voc	1.186 V	Voc	1.189 V
Pmax	2.032 mW	Pmax	2.036 mW
Ipm	1.980 mA	Ipm	1.981 mA
Vpm	1.026 V	Vpm	1.028 V
FF	83.34 %	FF	83.24 %
Eff	25.33 %	Eff	25.39 %
Dirr.	100 mW/cm^2	Dirr.	100 mW/cm^2
Mirr.	100 mW/cm^2	Mirr.	100 mW/cm^2
Scan Mode	Isc to Voc	Scan Mode	Voc to Isc

Ref. Device No. AK-200
Cal. Val. of Ref. 128.1mA at 100mW/cm^2

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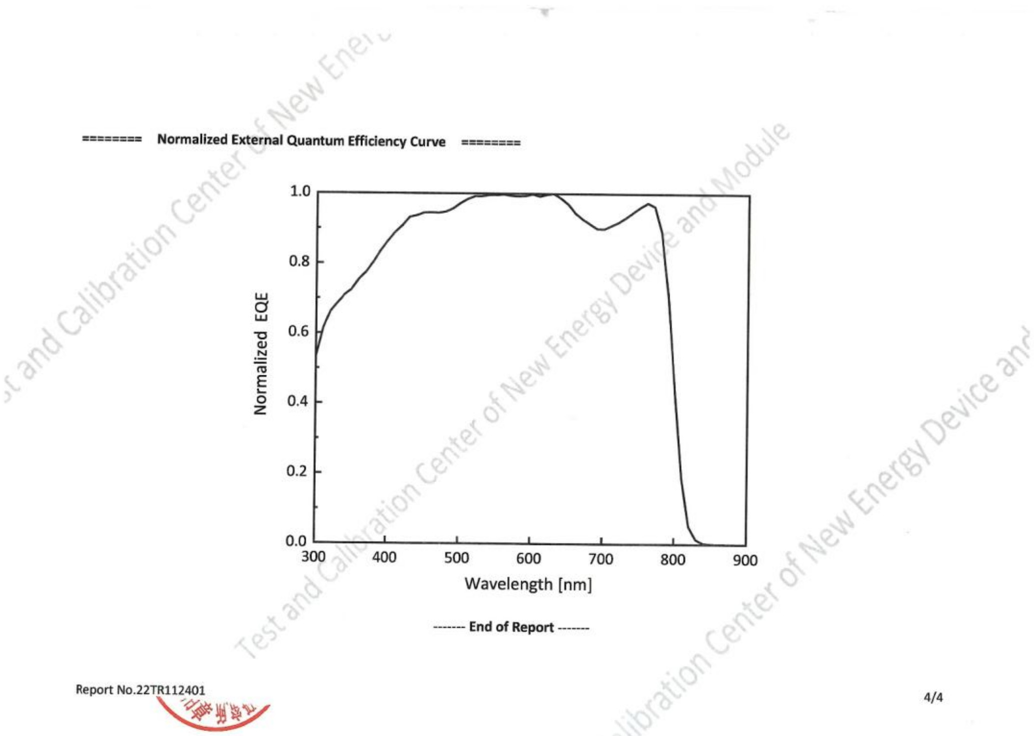
======== Normalized External Quantum Efficiency Curve ========

------- End of Report -------

Report No.22TR112401

4/4

Figure S31. Certification report of the target device based on the Perovskite(DMAcPA) by Shanghai Institute of Microsystem and Information Technology (*SIMIT*).

Figure S32 Device performances of the devices based on MeO-2PACz

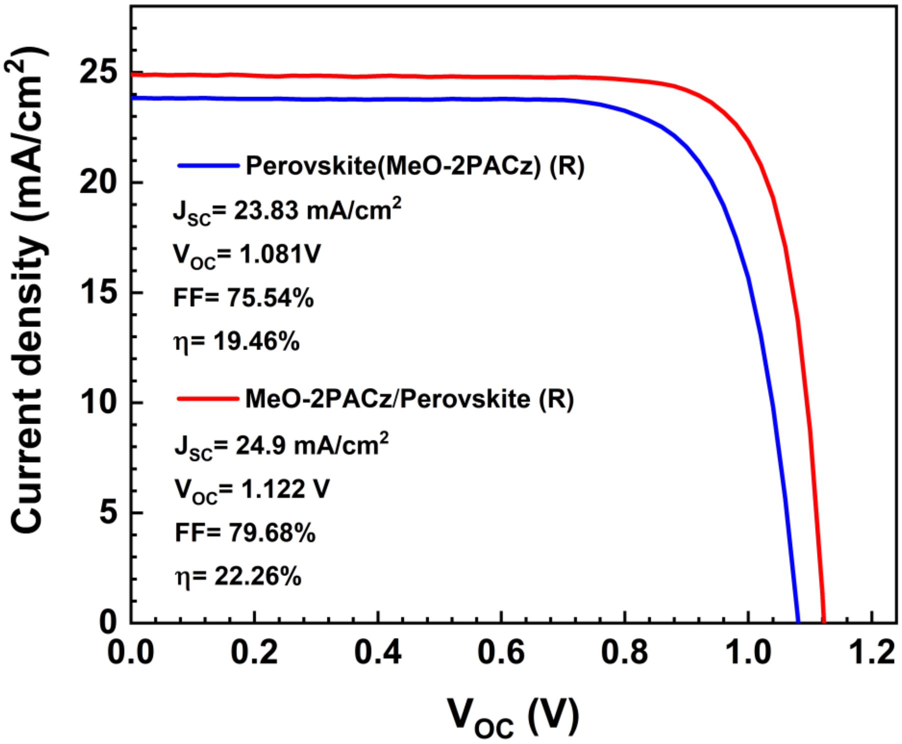

Figure S32. J-V curves (reverse scan) and photovoltaic performance parameters of the champion PSCs with ITO/Perovskite(MeO-2PACz)/$PC_{61}BM$/BCP/Ag and ITO/MeO-2PACz/Perovskite(undoped)/$PC_{61}BM$/BCP/Ag structures. The champion PCE of 19.46% was achieved (reverse scan) with a J_{SC} of 23.83 mA/cm^2, a Voc of 1.081 V and an FF of 75.54% for device based on MeO-2PACz doped perovskite. While the champion PCE of device based on the structure of ITO/MeO-2PACz/Perovskite is 22.26% with a J_{SC} of 24.9 mA/cm^2, a V_{OC} of 1.122 V and an FF of 79.68%.

Figure S33 Optimization of the dosage of MeO-2PACz in the doped Perovskite(MeO-2PACz)

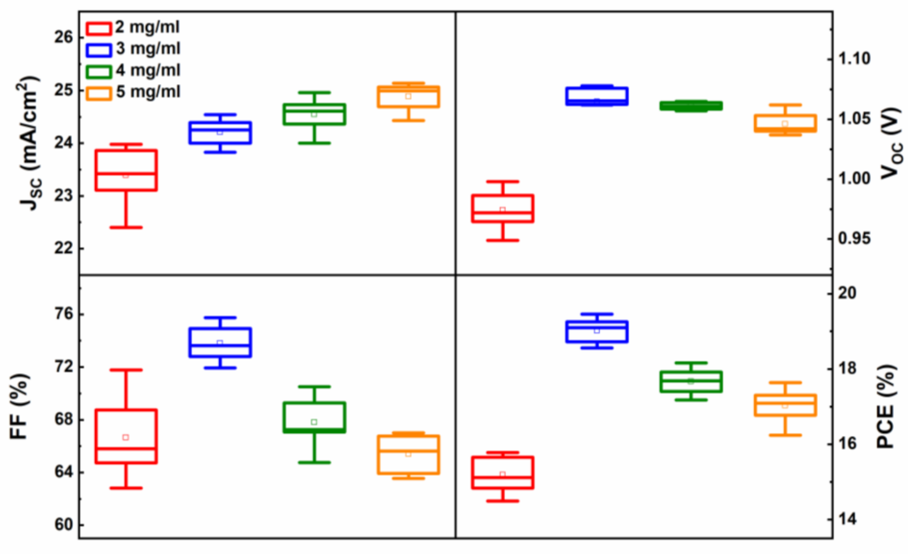

Figure S33. Statistic photovoltaic parameters of the target device based on the Perovskite(MeO-2PACz) in a batch of 24 devices, with different dosage of MeO-2PACz in perovskite precursor solution ranging from 2 mg/ml to 5 mg/ml. The peak PCE is determined by the highest FF and V_{OC} at 3 mg/ml.

Figure S34 Development summary of so called “HTL-free” PSCs

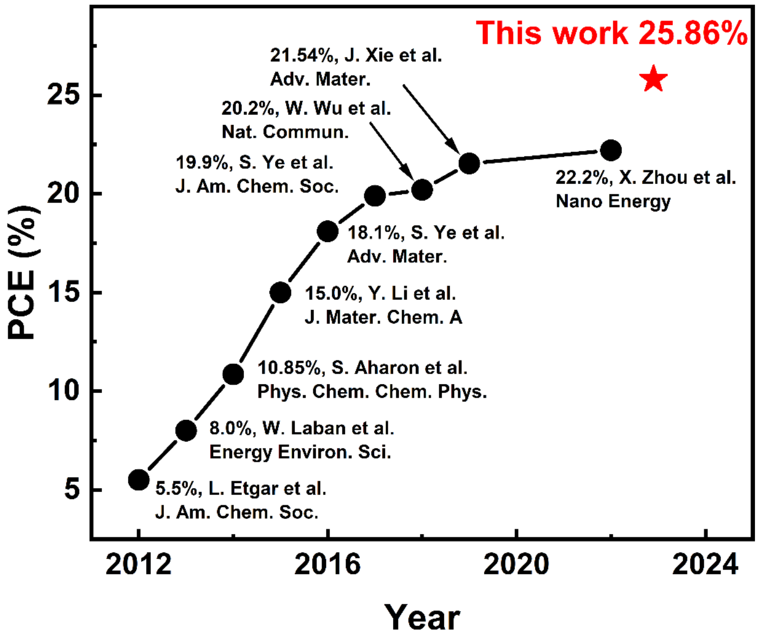

Figure S34. The PCE progress of so called “HTL-free” PSCs.(*9*-*17*)

Figure S35 Optimization of the dosage of DMAcPA in the doped perovskite(DMAcPA)

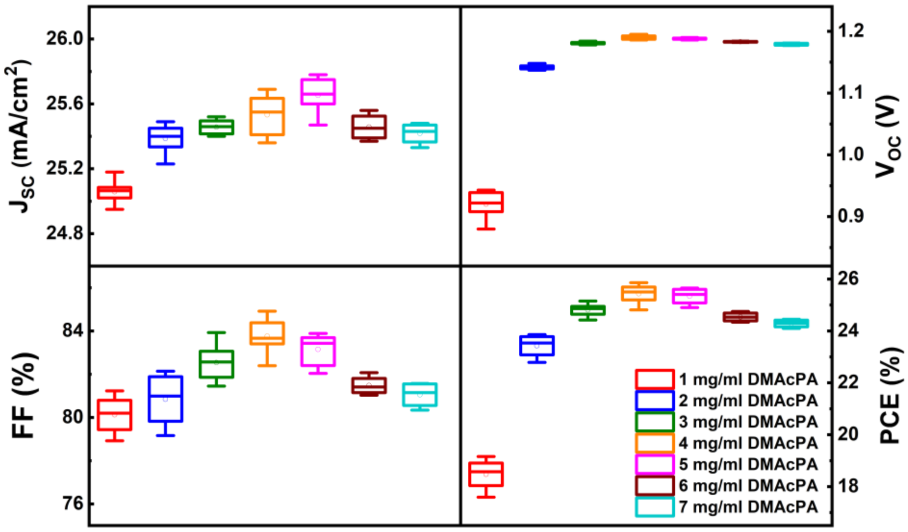

Figure S35. Statistic photovoltaic parameters of the target device based on the Perovskite(DMAcPA) in a batch of 24 devices, with different dosage of DMAcPA in perovskite precursor solution ranging from 1 mg/ml to 7 mg/ml. The centre line represents the median value, the small circle indicates the mean value, the box gives the standard deviation, and the whiskers represent the minimum and maximum values. The champion device comes from the optimal dosage of DMAcPA in the perovskite precursor solution after intensive optimization successively from 1 to 7 mg/ml (Fig. S28). For 1 mg/ml, both V_{OC} of 0.92 V and J_{SC} of 25.06 mA/cm^2 means the hole-extraction at perovskite/ITO interface is not effective, due to insufficient extruded interlayer according to that molecule-extrusion model. When the dosage rises to 2 mg/ml and more, V_{OC} jumps to over 1.1V and keeps a narrow distribution around 1.18V while J_{SC} also maintains from 25.2 to 25.8 mA/cm^2, which claims a broad window of DMAcPA dosage for high-quality perovskite film and perovskite/ITO interface. The peak PCE is determined by the highest FF at 4 mg/ml, and the dosage dependent FF variation reflects the extruded layer tends to be electrically compact for both hole-extraction and electron-blocking when the dosage increases from 2 mg/ml to 4 mg/ml while the further dosage increment would augment that extruded layer thickness and herein increase the series resistance and decline FF.

Figure S36 Transient absorption spectroscopy of the three kinds of perovskite film

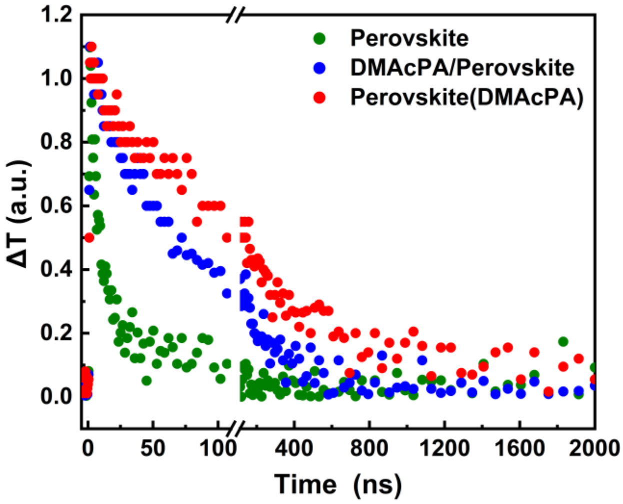

Figure S36. Transient absorption (TA) decay kinetics of ITO/Perovskite (green), ITO/DMAcPA/Perovskite (blue) and ITO/Perovskite(DMAcPA) (red) probed at 490 nm,in a transmission mode.

Figure S37 Initial device performances of the two devices for MPPT tests

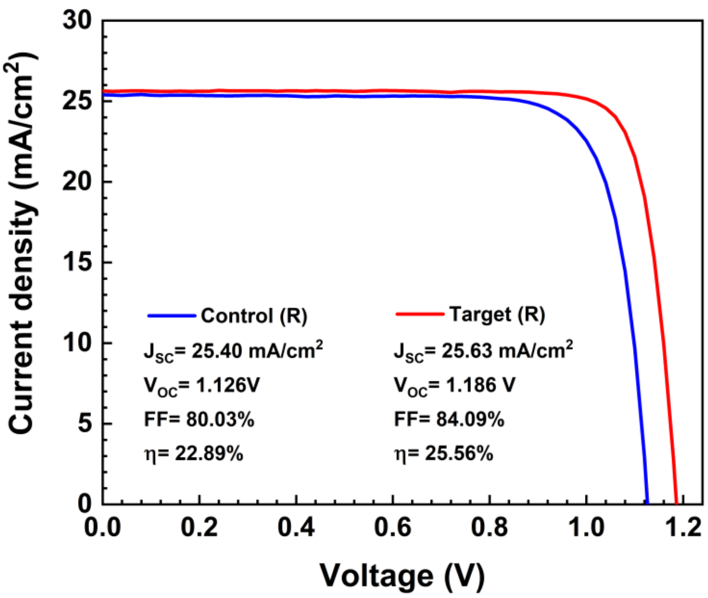

Figure S37 The initial J-V curves of the Control and Target PSCs tested before their MPP tracking.

Figure S38 Equivalent-circuit model ECM for EIS fitting

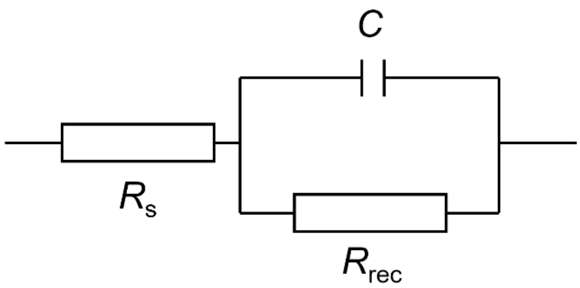

Figure S38. The equivalent-circuit model (ECM) employed for EIS fitting of PSCs, consist of the series resistance (R_s), the recombination resistance (R_{rec}) and the capacitance.

Table S7 Fitting parameters of PSCs from Nyquist plots.

	Rs (Ω)	**Rrec (Ω)**
Control	17.85	2046
Target	16.78	4028

Table S8 Deep-level traps parameters of the control and target devices.

	Traps	***E*t (eV)**	***σ* (cm²)**	***N*t (cm⁻³)**
Control	H1	Ev+0.383	5.93×10^{-16}	1.01×10^{14}
	H2	Ev+0.524	1.13×10^{-17}	1.81×10^{14}
Target	H1	*E*v+0.359	7.35×10^{-17}	3.27×10^{13}
	H2	*E*v+0.519	1.62×10^{-17}	1.10×10^{14}

References

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